

Explainable AI framework for digital twin-empowered battlefield threat assessment

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ARTICLE INFO

Keywords:

Digital twin
Explainable artificial intelligence
Deep learning
Military
Threat assessment

ABSTRACT

Artificial Intelligence (AI) is increasingly integrated into defense systems for real-time threat assessment and decision support. However, operational adoption remains limited due to concerns over transparency and trust. This study proposes an *Explainable AI (XAI)* framework for *Digital Twin (DTW)*-empowered battlefield threat assessment, combining multimodal deep learning and interpretable analytics. Heterogeneous data sources, including aerial imagery, acoustic signals, and sensor telemetry, are processed through unimodal modules (*Visual_{Deep}* and *Acoustic_{Deep}*) and fused via a DTW-based integration layer. Interpretability is achieved using SHAP (Shapley Additive exPlanations) for global and mission-specific insights. The experimental simulation results show that the proposed framework achieved 97.06% accuracy, 97.33% precision, 95.25% recall, and 97.29% specificity, outperforming state-of-the-art baselines. Moreover, SHAP analysis identified key threat indicators and sensor-level influences, enhancing situational awareness and operational trust.

1. Introduction

Modern warfare increasingly relies on Artificial Intelligence (AI) for rapid situational awareness, predictive analytics, autonomous mission planning, and adaptive threat assessment [1]. Contemporary battlefield environments are highly dynamic, uncertain, and data-intensive, generating massive volumes of heterogeneous information from unmanned aerial vehicles (UAVs), radar systems, IoT-enabled defense infrastructures, satellite surveillance, acoustic sensors, cyber-intelligence platforms, and edge computing nodes [2]. The growing complexity and scale of these multimodal battlefield data streams have accelerated the adoption of AI-driven defense analytics for low-latency operational decision-making and mission-critical intelligence support. Despite significant advances in AI-enabled military intelligence systems, several critical challenges continue to hinder their reliable deployment in combat environments. One major challenge is the heterogeneity and high dimensionality of battlefield data, where information acquired from multiple sensing modalities often exhibits inconsistent temporal resolution, varying noise characteristics, missing observations, and asynchronous acquisition patterns. Efficiently integrating such heterogeneous multimodal information into a unified operational representation remains a difficult task in battlefield intelligence systems. Another major challenge arises from the dynamic and continuously evolving nature of battlefield environments. Operational conditions

rapidly change due to adversarial movement, environmental disturbances, communication disruptions, sensor failures, and unpredictable tactical events. Conventional AI models frequently struggle to maintain stable prediction performance under such non-stationary and uncertain operational conditions, particularly in distributed multi-domain battlefield environments. Furthermore, existing battlefield intelligence systems often lack sufficient context-awareness and adaptive synchronization capabilities. Traditional surveillance and threat assessment frameworks generally process sensory inputs independently without maintaining continuous interaction between the physical battlefield and its virtual representation. Consequently, these systems are often unable to provide synchronized situational awareness, predictive mission simulation, or adaptive tactical decision support. Recent advances in Digital Twin (DTW)-enabled battlefield intelligence systems have introduced new opportunities for building adaptive and data-centric defense ecosystems [3]. Unlike conventional simulation frameworks, DT systems maintain continuous bidirectional interaction between physical battlefield assets and their cyber representations through real-time sensor-driven synchronization. By integrating live battlefield telemetry from UAVs, radar platforms, wearable soldier devices, edge-IoT nodes, communication infrastructures, and surveillance systems, DTW frameworks enable predictive mission simulation, adaptive mission orchestration, anomaly detection, and proactive threat mitigation [4].

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<https://doi.org/10.1016/j.array.2026.100986>

Received 22 April 2026; Received in revised form 1 June 2026; Accepted 5 June 2026

Available online 11 June 2026

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However, despite their advantages, several important limitations remain in existing DT-enabled defense systems. Current DTW frameworks often face difficulties in:

- handling large-scale heterogeneous multimodal data streams,
- maintaining low-latency synchronization under operational constraints,
- adapting to uncertain and adversarial battlefield conditions,
- providing scalable distributed intelligence across multi-domain environments,
- ensuring trustworthy and interpretable AI-driven decision-making.

Among these challenges, the lack of explainability and operational transparency remains one of the most critical barriers to practical deployment. Most conventional deep learning models operate as black-box systems, generating highly accurate predictions without clearly explaining how specific battlefield features influence threat identification or tactical recommendations [5]. In mission-critical defense operations, such opaque decision-making mechanisms reduce operational trust, accountability, and human oversight capability. Military commanders and defense analysts require interpretable reasoning to validate AI-driven threat predictions, particularly in high-risk operational scenarios involving adversarial uncertainty and ethical decision-making constraints [6].

1.1. Research domain

Recent developments in Machine Learning (ML) and Deep Learning (DL) have significantly advanced intelligent defense and battlefield analytics applications, including autonomous surveillance, target recognition, cyber-defense, signal intelligence, tactical communication optimization, and adaptive navigation systems [7]. Traditional ML techniques generally rely on handcrafted feature engineering and often struggle to efficiently process large-scale heterogeneous battlefield data. In contrast, DL architectures automatically learn complex spatial, temporal, and contextual patterns from multimodal sensory environments [8]. Despite their strong predictive capabilities, most DL models remain difficult to interpret, thereby limiting their adoption in mission-critical defense systems where transparency, accountability, and human-centered decision support are essential [9]. To address the limitations of black-box deep learning models, recent advances in Explainable Artificial Intelligence (XAI) have demonstrated considerable potential for improving transparency and interpretability in intelligent healthcare and diagnostic systems. Existing studies have successfully utilized XAI-integrated deep learning frameworks for diabetes prediction, Alzheimer's disease analysis, lung cancer detection, and lung adenocarcinoma classification using interpretable feature learning, attention mechanisms, and optimization-driven architectures [8,9]. These studies highlighted the importance of integrating predictive intelligence with explainability to improve reliability, trustworthiness, and human interpretability in high-stakes decision-making environments. Furthermore, hybrid architectures combining convolutional neural networks, transformer-based models, optimization algorithms, and explainability mechanisms have demonstrated improved performance in complex classification and prediction tasks. The success of XAI-driven frameworks in critical healthcare applications motivates their adaptation to battlefield intelligence systems, where transparent threat reasoning and interpretable tactical recommendations are equally essential. In battlefield threat assessment, multimodal data sources such as visual imagery, acoustic signatures, telemetry streams, radar signals, and environmental sensing provide complementary perspectives of evolving operational threats [10]. Reliance on a single sensing modality often results in incomplete situational awareness, reduced prediction reliability, and poor adaptability under uncertain combat conditions. Therefore, integrating multimodal information within a DT-enabled

framework can provide a more comprehensive, adaptive, and context-aware representation of battlefield dynamics [11]. Nevertheless, effective operational deployment requires AI systems that not only achieve high predictive accuracy but also provide interpretable explanations regarding detected threats, mission risks, synchronization behavior, and adaptive control decisions [12]. Motivated by these challenges, this study proposes an XAI-driven DTW-enabled battlefield intelligence framework that integrates multimodal deep learning, cyber-physical synchronization, and interpretable battlefield analytics for transparent and adaptive threat assessment. The proposed framework aims to address the limitations of existing battlefield intelligence systems by improving multimodal fusion, operational synchronization, contextual awareness, explainability, and adaptive decision support. By integrating deep learning, explainable AI, and DTW synchronization mechanisms, the proposed framework enhances operational trust, predictive situational awareness, adaptive threat reasoning, and mission-critical battlefield analytics.

1.2. Research objectives and contributions

Although AI has demonstrated remarkable success in defense analytics, its limited interpretability and lack of transparent decision-making continue to hinder operational trust and large-scale deployment. XAI provides interpretable and mission-aware insights that are essential for improving situational awareness, accountability, and confidence in autonomous defense systems. Motivated by these challenges, this study establishes the following research objectives:

1. Develop multimodal deep learning architectures (*Visual-Deep* and *Acoustic-Deep*) for detecting and classifying battlefield threats using heterogeneous data sources, including aerial imagery, acoustic signals, and sensor telemetry streams.
2. Design a Digital Twin-enabled integration mechanism that dynamically fuses multimodal outputs to represent evolving battlefield states and enhance threat severity assessment.
3. Integrate SHAP (SHapley Additive exPlanations) for global and mission-specific interpretability, thereby ensuring transparent and explainable decision support.
4. Evaluate the proposed framework against state-of-the-art benchmark models in terms of detection accuracy, robustness, scalability, and interpretability under simulated combat conditions.

Fig. 1 presents the conceptual data flow of the proposed *XAI Framework for DTW-empowered Battlefield Threat Assessment*. The framework establishes an intelligent architecture for adaptive mission assurance and real-time threat evaluation. By integrating IoT-based sensing, DTW-driven analytics, and XAI, it enables predictive situational awareness and dynamic decision optimization to enhance mission resilience and operational efficiency.

Paper organization. Section 2 reviews recent advancements in AI-driven defense analytics for battlefield intelligence. Section 3 details the architecture and core components of the proposed explainable threat assessment framework. Section 4 presents experimental evaluations validating the model's performance and interpretability. Finally, Section 5 concludes with key findings and outlines potential directions for future research.

2. Related work

Recent advancements in AI, DL, XAI, and distributed intelligence have significantly influenced intelligent defense analytics and autonomous battlefield decision-making systems. Although existing studies have demonstrated substantial progress in predictive intelligence, surveillance automation, and interpretable learning, several

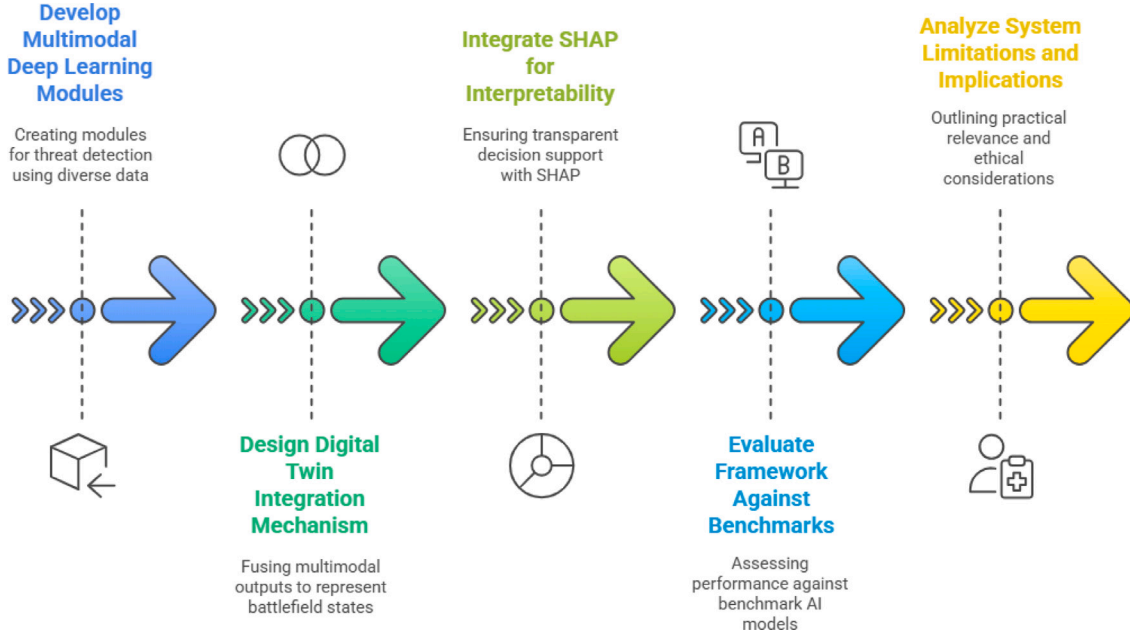


Fig. 1. Conceptual data workflow.

Table 1

Comparative analysis of existing battlefield intelligence frameworks and the proposed XAI-DT framework.

Ref.	Modalities used	Core approach	XAI support	DTW integration	Real-time context awareness	Research gap/limitation
[13]	Aerial imagery	CNN-based object detection	No	No	Limited	Focused on isolated visual target detection; lacks multimodal fusion, explainability, and adaptive battlefield synchronization.
[14]	Surveillance and defense analytics data	ML/DL survey framework	Partial	No	No	Provides a survey-level discussion without proposing an operational explainable or DTW-enabled framework.
[15]	Tactical imagery	Federated deep learning	No	No	Limited	Primarily focused on privacy preservation; lacks multimodal intelligence fusion and interpretable decision reasoning.
[16]	Multi-source defense data	Federated learning + blockchain	No	No	Limited	Addresses secure distributed learning but lacks explainable AI mechanisms and real-time DTW synchronization.
[17]	UAV imagery	Collaborative federated learning	No	No	Partial	Limited to vision-based analysis; lacks multimodal adaptability and operational transparency.
[18]	Combat imaging data	Personalized federated learning	No	No	Limited	Improves distributed training efficiency but does not provide contextual threat reasoning or explainability.
[19]	UAV and satellite data	Adaptive federated averaging (AdaFed)	No	No	Partial	Focused on aggregation optimization; lacks SHAP-based interpretability and multimodal battlefield modeling.
[20]	Multi-sensor defense data	Reinforcement learning + transfer learning	Partial	No	Partial	Enhances adaptive learning but lacks Digital Twin synchronization and interpretable multimodal reasoning.
[21]	Distributed sensor networks	Multi-site federated framework	No	No	Partial	Emphasizes distributed learning efficiency without contextual DTW integration or explainable threat analytics.
[22]	Surveillance video feeds	Lightweight CNN framework	No	No	Limited	Unimodal architecture with limited contextual awareness and no explainability support.
[23]	Mobility and environmental data	Crowd-intelligence analytics	Partial	No	Partial	Focused on behavioral analytics; lacks multimodal battlefield synchronization and adaptive threat explainability.
This work	Visual + Acoustic + Sensor Telemetry	Multimodal deep learning + SHAP + DTW synchronization	Yes	Yes	Yes	Provides interpretable multimodal threat assessment through continuous Digital Twin synchronization, adaptive battlefield intelligence, and explainable AI-driven decision support.

methodological limitations remain regarding multimodal fusion, operational synchronization, explainability, and adaptive battlefield reasoning. Nagaraj et al. [24] utilized XAI techniques to identify critical diabetic attributes and improve prediction transparency through interpretable rule mining and decision analysis. The study evaluated multiple machine learning models and selected the optimal framework based on classification accuracy. The proposed system incorporated conventional clinical parameters, including glucose level, BMI, age, and insulin, together with external risk factors to enhance diabetes prediction performance. Similarly, Sudar et al. [25] investigated Alzheimer's disease stage identification using XAI-driven medical image analysis. Their framework employed Layer-wise Relevance Propagation (LRP) to provide feature-level interpretability, while deep learning architectures such as Convolutional Neural Networks (CNNs) and VGG-16 were utilized to improve classification accuracy and diagnostic reliability. Robert et al. [26] proposed an advanced DL- and XAI-enabled framework for automated lung cancer prediction using CT images. The framework employed a Fast Local Laplacian Filter (FLLF) during preprocessing to preserve edge information and image contrast. For segmentation, the study integrated Fuzzy C-Means (FCM) clustering with Markov Random Fields (MRF) to improve lesion localization and spatial coherence. Morphological features were extracted using contour-based descriptors, while classification was performed through a Pyramid Vision Transformer (PVT)-ResNet architecture that combined convolutional learning with global attention mechanisms for improved diagnostic accuracy. Likewise, Lakshmi et al. [27] introduced a DL-XAI framework for lung adenocarcinoma classification using CT images. The study performed intensity normalization and volume cropping during preprocessing, followed by data augmentation using MedGAN. For classification, the authors developed an Optimal NAS-MoE architecture integrating NASNet with a Mixture-of-Experts (MoE) Transformer module to classify lung adenocarcinoma into Atypical Adenomatous Hyperplasia (AAH), Adenocarcinoma in Situ (AIS), and Minimally Invasive Adenocarcinoma (MIA). Beyond healthcare analytics, several studies have explored AI-enabled frameworks for intelligent surveillance and military situational awareness. Maheriya et al. [13] proposed a CNN-based architecture for automated object recognition in aerial surveillance, achieving classification accuracies between 89.2% and 98.6%. Bhatia et al. [14] presented a comprehensive survey of ML- and DL-based approaches for military situational awareness and battlefield intelligence, providing useful taxonomies and analytical insights. However, the study primarily focused on conventional learning models and did not address multimodal fusion or explainability-driven intelligence. Recent research has also emphasized federated and distributed learning frameworks to improve operational data security while utilizing geographically distributed battlefield datasets. Dwivedi et al. [28] conducted a comprehensive survey of XAI programming techniques and interpretability methodologies, categorizing major explainability paradigms and mapping them to widely adopted machine learning frameworks and software toolkits. Li et al. [29] proposed a hybrid coupling paradigm integrating model-driven low-rank representation with data-driven deep learning through disentangled prior learning. The framework effectively captured background priors for anomaly extraction from hyperspectral imagery, thereby improving anomaly detection performance. To support secure and distributed battlefield intelligence, Arora et al. [15] developed a federated deep learning framework for tactical image analysis using adaptive aggregation across distributed edge nodes. Alghamedy et al. [16] integrated blockchain with federated learning to enhance confidentiality and integrity in mission-critical communications. Similarly, Lyu et al. [17] proposed a collaborative UAV reconnaissance framework capable of addressing non-IID data distributions and communication instability in distributed operational environments. Zhang et al. [18] introduced a personalized federated learning model to optimize local training across heterogeneous combat datasets, while Wang et al. [19] improved

aggregation efficiency through adaptive weighting strategies. Furthermore, Oh et al. [20] employed reinforcement learning for dynamic node selection to mitigate data imbalance and resource constraints in distributed battlefield intelligence systems. Wang et al. [21] further proposed a multi-site federated framework with drift correction mechanisms for robust multi-domain threat identification. Complementary studies have focused on lightweight and edge-oriented battlefield intelligence architectures. Aljumah et al. [22] introduced a lightweight CNN-based communication framework emphasizing edge-level inference and secure data transmission for defense environments. Extending beyond conventional sensor analytics, Halboob et al. [23] proposed a crowd-intelligence framework that utilized mobility and environmental information to forecast potential conflict zones and support adaptive mission planning. Although these studies have contributed substantially to AI-enabled analytics, distributed learning, and interpretable intelligence, several important limitations remain. Most existing frameworks focus on isolated tasks such as medical diagnosis, object recognition, or federated optimization without addressing synchronized cyber-physical battlefield representation. Furthermore, many existing defense-oriented systems lack integrated multimodal fusion, real-time DTW synchronization, adaptive threat reasoning, and mission-specific explainability. In addition, current XAI approaches are often evaluated primarily in healthcare or generic AI domains rather than mission-critical combat environments characterized by adversarial uncertainty, heterogeneous sensing, and rapidly evolving operational dynamics. Therefore, there remains a significant research gap in developing a unified XAI-driven DTW-enabled battlefield intelligence framework capable of integrating multimodal sensory analytics, adaptive cyber-physical synchronization, interpretable threat assessment, and scalable distributed intelligence for transparent and mission-aware battlefield decision support.

2.1. Research gap

As summarized in Table 1, existing studies exhibit several important methodological and operational limitations:

1. Most existing frameworks primarily focus on isolated tasks such as target recognition, surveillance, anomaly detection, or tactical communication optimization, rather than providing a comprehensive and adaptive battlefield threat assessment framework.
2. A large proportion of current approaches rely on unimodal data sources, predominantly visual imagery or telemetry information, without effectively exploiting the complementary advantages of multimodal sensory fusion for robust situational awareness.
3. Existing AI-driven defense systems often provide limited interpretability and operational transparency, offering minimal insight into feature-level decision reasoning and thereby reducing user trust, accountability, and mission-level explainability.
4. Current battlefield intelligence frameworks rarely incorporate synchronized cyber-physical interaction through Digital Twin-enabled architectures, limiting their capability for adaptive battlefield modeling, predictive simulation, and real-time operational synchronization.
5. Most existing XAI-based studies are evaluated in healthcare or general AI application domains, with limited investigation of explainability mechanisms in mission-critical and adversarial battlefield environments characterized by uncertainty, heterogeneous sensing, and rapidly evolving operational conditions.

To address these limitations, the present study proposes a multimodal XAI-driven DTW-enabled battlefield intelligence framework that integrates visual imagery, acoustic signals, and sensor telemetry within a synchronized cyber-physical architecture. The proposed framework combines multimodal deep learning, adaptive DTW synchronization, and SHAP-based interpretability to improve threat detection, operational transparency, contextual awareness, and mission-critical decision

support. By integrating explainable intelligence with multimodal battlefield analytics, the framework provides a more reliable, adaptive, and interpretable alternative to existing unimodal and non-explainable defense AI systems.

2.1.1. Comparative motivation and research significance

Table 1 highlights that most existing battlefield intelligence frameworks primarily focus on isolated objectives such as federated learning, visual target detection, distributed optimization, or secure communication. Although these approaches improve specific components of intelligent defense analytics, they still exhibit several fundamental limitations in the context of modern battlefield intelligence requirements. A key limitation of existing approaches is the lack of explainability mechanisms, as most conventional deep learning-based defense systems operate as black-box models. These models provide high predictive performance but fail to offer interpretable reasoning for threat predictions or tactical recommendations, thereby limiting operational trust, transparency, and human-in-the-loop decision validation. In contrast, XAI techniques such as SHAP-based feature attribution and visual saliency mechanisms enable fine-grained interpretation of modality-specific contributions, which is critical for mission-critical defense applications. Another major limitation is the absence of real-time DTW synchronization. Most existing frameworks process multimodal data in a static or loosely coupled manner without maintaining a continuously updated cyber-physical representation of the battlefield. As a result, they are unable to capture real-time state evolution, feedback-driven adaptation, and predictive simulation of dynamic battlefield conditions. This restricts their applicability in rapidly changing and adversarial environments. Furthermore, existing studies often lack multimodal fusion and contextual battlefield awareness, as they typically rely on single-modality or weakly integrated data sources such as visual surveillance or isolated sensor streams. This limits their ability to capture complex cross-modal dependencies required for robust threat assessment in heterogeneous operational scenarios. In contrast, the proposed XAI-DTW framework explicitly addresses these limitations by integrating multimodal battlefield intelligence, continuous DTW synchronization, adaptive situational awareness, and explainable AI mechanisms within a unified cyber-physical defense ecosystem. The incorporation of SHAP-based feature attribution, along with DTW-enabled real-time state synchronization, enables transparent, interpretable, and context-aware threat reasoning under dynamic battlefield conditions. Therefore, unlike conventional unimodal or black-box defense analytics, the proposed framework provides a scalable, interpretable, and context-aware multimodal intelligence architecture capable of supporting real-time, mission-critical defense decision-making with improved transparency, adaptability, and operational trustworthiness.

3. Proposed model

The proposed system introduces an XAI-driven framework for DTW-empowered battlefield threat assessment, integrating heterogeneous data modalities to enhance prediction accuracy, transparency, and operational reliability. As illustrated in Fig. 2, the framework comprises four key modules: (i) Multimodal data acquisition, (ii) Threat prediction, (iii) Dynamic integration, and (iv) Explainable decision analysis. Each module is detailed below.

3.1. Multimodal data acquisition module

This module collects and synchronizes diverse battlefield data streams, encompassing visual, acoustic, and sensor-based telemetry from distributed defense assets. The objective is to capture real-time situational information reflecting troop movements, environmental conditions, and potential hostile activity. Let X_s represent sensor telemetry, X_v denote visual imagery, and X_a refer to acoustic intelligence. Following standard multimodal data fusion principles [30], the

integrated dataset is represented as the union of heterogeneous data sources:

$$X_t = X_s \cup X_v \cup X_a \quad (1)$$

- 1. Acoustic Dataset:** Acoustic signals include gunfire, explosions, vehicle movements, and drone acoustic patterns captured via distributed IoMT-based field sensors. Data were compiled from open defense sound repositories and simulated combat environments to ensure diversity in signal type and background context. Preprocessing involved denoising, segmentation, and normalization under a signal-to-noise ratio (SNR) threshold of 20 dB. Samples were annotated as *threat-positive* (hostile or combat-related) or *threat-negative* (ambient or non-combat) based on ground-truth event metadata. Data augmentation — via pitch shifting, time-stretching, and ambient mixing — expanded the dataset while maintaining acoustic fidelity for operational environments.
- 2. Visual Dataset:** The visual modality comprises aerial and ground imagery obtained from UAV surveillance, reconnaissance satellites, and tactical cameras. Datasets were sourced from defense simulation platforms and verified using open-source intelligence imagery. Inclusion criteria ensured high-resolution data (minimum 720p) with confirmed threat annotations, while exclusion criteria filtered out corrupted or ambiguous samples. Preprocessing involved image resizing, histogram equalization, and environmental normalization (lighting and weather conditions). Augmentation was applied using geometric and spectral transformations to enhance model generalization. The resulting dataset included over 160,000 frames representing varied terrain, threat types, and engagement conditions.
- 3. Sensor Telemetry:** Complementary IoT-based telemetry data — such as vibration, infrared, and motion sensor readings — were synchronized with acoustic and visual inputs to provide contextual awareness. This sensor fusion enables the DTW to construct an evolving virtual representation of the battlefield, ensuring high-fidelity alignment between the physical and digital environments. The fused dataset forms the foundation for adaptive threat modeling and explainable decision support.

Preprocessing pipeline: The multimodal data preprocessing workflow consisted of four sequential operations — filtering, segmentation, augmentation, and normalization — ensuring consistency and robustness across sensor, acoustic, and visual streams.

- 1. Filtering:** A sliding median filter (γ) was used to eliminate high-frequency interference and sensor jitter. Following standard signal preprocessing and threshold-based event detection methods [31], discrete event labels $y_t(e)$ were binarized using a hard-threshold rule:

$$\bar{y}_t(e) = \begin{cases} 1, & y_t(e) > \theta_b \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

This process suppresses transient anomalies and ensures a clean event-level signal representation.

- 2. Segmentation:** Audio signals were divided into 30–35 ms frames with a 5 ms overlap, whereas UAV images were standardized to 224×224 pixels. This segmentation ensured alignment between temporal and spatial modalities for synchronized feature extraction.
- 3. Augmentation:** To enhance generalization and mitigate overfitting, domain-appropriate augmentation was applied to all modalities. For acoustic data, transformations included pitch shifting (± 2 semitones), time-shifting (up to 60 ms), and additive noise within an SNR range of 20–30 dB. For visual data, random rotations ($\pm 15^\circ$), scaling (0.9–1.1 $\times$), and horizontal mirroring were applied to simulate dynamic operational perspectives. Augmentation parameters were tuned to approximately double the

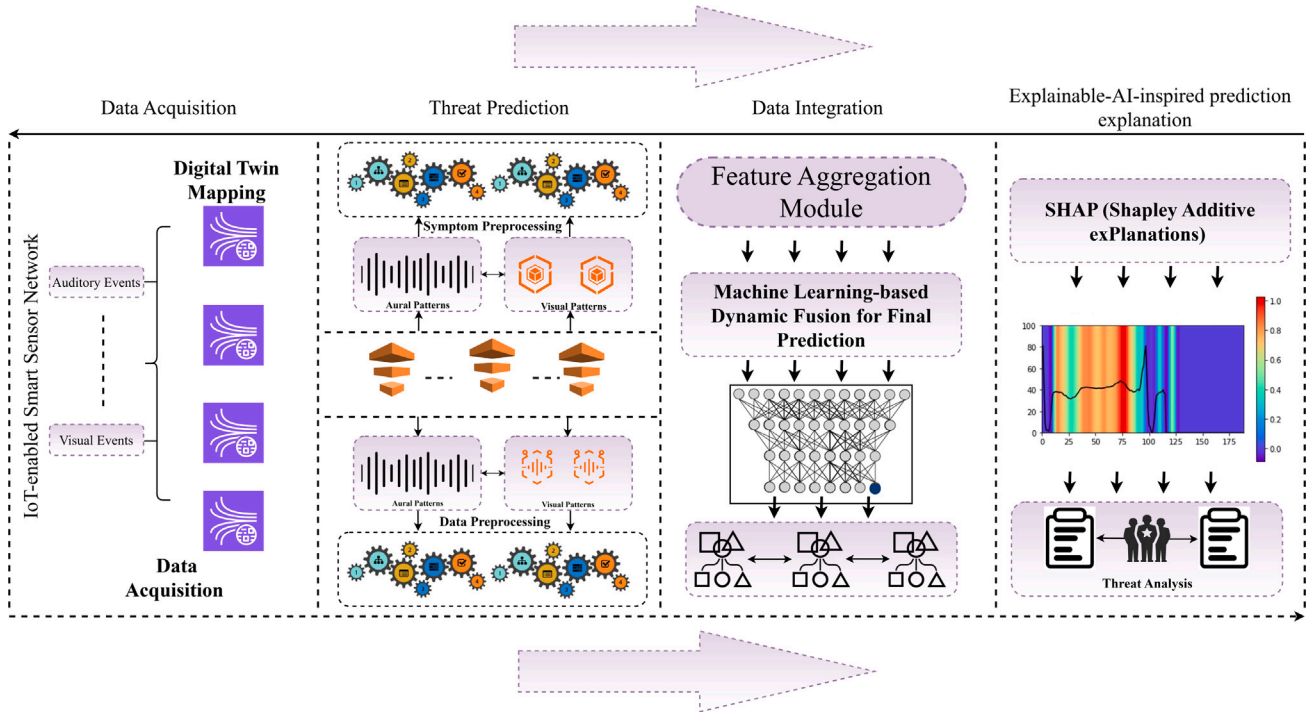


Fig. 2. Proposed model.

Table 2
Multimodal dataset characteristics, distribution, and augmentation specifications.

Data source	Category/Type	Raw samples	Augmented samples	Total samples	Augmentation techniques
Acoustic (IoT sensors)	Communication logs	817	1428	1695	Time-shift, Gaussian noise injection Pitch shift, temporal stretching Frequency masking, noise filtering Spectral augmentation, time warping
	Weapon impact audio	848	1278	1576	
	Vehicle movement	986	1388	1824	
	Drone propulsion	1072	1492	2014	
Visual (UAV imaging)	Tactical frames	~40,150	161,408	~201,008	Rotation, scaling, mirroring

dataset volume while maintaining semantic and operational fidelity. Balanced augmentation ensured uniform class expansion and reduced bias.

4. **Normalization:** Feature scaling was performed to stabilize model training. Audio features (e.g., MFCCs) were z-score normalized to zero mean and unit variance, while visual pixel intensities were rescaled to the $[0, 1]$ interval. This normalization balanced modality contributions during DT-driven learning and fusion.

Table 2 provides an overview of dataset composition, highlighting the diversity of acoustic and visual samples, the augmentation strategy employed, and the resulting total data volume integrated within the Explainable DTW environment.

3.1.1. DTW Modulation Stage (DTMS)

The *Digital Twin Modulation Stage (DTMS)* establishes a continuously synchronized cyber-physical representation of the operational battlespace by integrating real-time battlefield sensing, adaptive state modeling, predictive simulation, and feedback-driven control. Unlike a conventional state-space framework that only models system evolution mathematically, the proposed DTMS incorporates continuous bidirectional interaction between the physical battlefield and its virtual counterpart, thereby enabling real-time situational awareness, predictive analysis, and adaptive mission orchestration. Let the operational battlefield state at time t be represented as:

$$\mathbf{x}(t) = [x_1(t), x_2(t), \dots, x_n(t)]^T \quad (3)$$

where each state variable $x_i(t)$ corresponds to mission-critical battlefield attributes such as troop mobility, platform health, communication quality, environmental dynamics, and adversarial activities. The virtual DTW state evolves according to:

$$\mathbf{x}_{DT}(t+1) = f(\mathbf{x}_{DT}(t), \mathbf{u}(t), \mathbf{w}(t)) \quad (4)$$

where $f(\cdot)$ denotes nonlinear battlefield dynamics, $\mathbf{u}(t)$ represents mission control actions, and $\mathbf{w}(t)$ captures stochastic uncertainties and environmental perturbations.

Real-Time Physical-Virtual Synchronization: A key characteristic distinguishing the DTW from a conventional simulation model is the continuous synchronization between the physical battlefield and the virtual twin. Real-time multimodal sensor streams collected from UAVs, IoT-enabled military assets, radar systems, wearable devices, and surveillance infrastructures are continuously assimilated into the DTW model. Let the incoming battlefield observation vector be:

$$\mathbf{z}(t) = [z_1(t), z_2(t), \dots, z_m(t)]^T \quad (5)$$

The synchronization mechanism updates the DTW state using sensor-driven data assimilation:

$$\mathbf{x}_{DT}(t) \leftarrow \alpha \mathbf{x}_{DT}(t) + (1 - \alpha) \mathbf{x}_{phys}(t) \quad (6)$$

where:

- $\mathbf{x}_{phys}(t)$ denotes real-time physical battlefield measurements,
- $\alpha \in [0, 1]$ controls the synchronization and adaptation rate.

To further reduce state deviation between the physical and virtual domains, the synchronization error is defined as:

$$e(t) = \|\mathbf{x}_{phys}(t) - \mathbf{x}_{DT}(t)\|_2 \quad (7)$$

The DTW continuously minimizes $e(t)$ through iterative sensor-driven state refinement, thereby maintaining high-fidelity alignment between the operational battlefield and its digital representation.

Bidirectional Interaction and Adaptive Feedback Control: In contrast to conventional predictive models, the DTW not only receives data from the physical battlefield but also actively influences battlefield operations through adaptive feedback mechanisms. The DTW generates mission recommendations, threat alerts, and optimized control strategies that are transmitted back to operational units. The adaptive control policy is formulated as:

$$\mathbf{u}(t) = \pi(\mathbf{x}_{DT}(t)) \quad (8)$$

where $\pi(\cdot)$ denotes the DT-driven decision policy responsible for tactical adaptation, threat mitigation, and mission optimization. The generated control actions may include:

- dynamic troop reallocation,
- autonomous UAV trajectory adjustment,
- adaptive communication rerouting,
- threat-aware resource prioritization,
- predictive maintenance scheduling.

Thus, the DTW forms a closed-loop cyber-physical feedback architecture in which battlefield observations continuously update the digital twin, while the twin simultaneously influences physical mission execution.

Probabilistic Multi-Scenario Twin Generation: To model uncertainty across diverse combat environments, the DTMS constructs an ensemble of probabilistic battlefield twins using Monte Carlo simulation:

$$\mathbf{x}_{DT}^j(t) \sim P(\mathbf{x}(t) | D), \quad j = 1, 2, \dots, M \quad (9)$$

where:

- M represents the number of operational twin realizations,
- D denotes observed multimodal battlefield data.

This probabilistic DTW ensemble enables predictive evaluation of multiple operational trajectories, adversarial behaviors, and mission risk conditions under uncertain battlefield dynamics.

Integrated DTW Representation: The complete DTW representation is expressed as:

$$I(DTW) = [I_{data}, I_{TD}, I_{OD}, I_{SD}, Out] \quad (10)$$

$$I_{data} = \{d_i\}, \quad i = 1, \dots, k \quad (11)$$

$$I_{TD} = g_{tact}(\mathbf{x}_{DT}(t)) \quad (12)$$

$$I_{OD} = g_{oper}(\mathbf{x}_{DT}(t)) \quad (13)$$

$$I_{SD} = g_{sens}(\mathbf{x}_{DT}(t)) \quad (14)$$

$$Out = h(I(DTW)) \quad (15)$$

where:

- I_{data} : multimodal battlefield telemetry,
- I_{TD} : tactical intelligence descriptors,
- I_{OD} : operational mission indicators,
- I_{SD} : sensor and platform diagnostics,
- Out : predictive threat alerts, adaptive mission recommendations, and optimized battlefield control actions.

Therefore, the proposed DTMS extends beyond a conventional state-space or simulation model by incorporating continuous physical-virtual

synchronization, real-time sensor-driven state assimilation, probabilistic predictive modeling, and bidirectional adaptive feedback control. These capabilities collectively establish the framework as a true Digital Twin-enabled battlefield intelligence system capable of real-time monitoring, predictive situational analysis, and autonomous operational adaptation.

3.2. Threat prediction module

The Threat Prediction Module serves as the analytical core of the proposed framework, employing modality-specific models — $Acoustic_{Deep}$ and $Visual_{Deep}$ — to identify, classify, and interpret potential battlefield threats from heterogeneous data streams. These models operate in parallel to capture both temporal (acoustic) and spatial (visual) threat patterns before integration through the XAI layer.

3.2.1. Acoustic threat analysis

The $Acoustic_{Deep}$ network is tailored to process mission-critical auditory data such as weapon discharge, drone propulsion, and vehicle movement signatures. It combines Mel Frequency Cepstral Coefficients (MFCC)-based feature extraction with a hybrid CNN-GRU architecture, enabling both spatial and temporal feature learning. The Mel-scale transformation, widely used in speech and audio signal processing [32], is defined as:

$$m = 2595 \times \log_{10} \left(1 + \frac{f}{700} \right) \quad (16)$$

The model comprises five convolutional layers, gated recurrent units (GRUs), and adaptive pooling layers with batch normalization and LeakyReLU activation. Subsampling is performed using L_p -norm pooling, a generalized pooling strategy commonly used in deep learning [33]:

$$L_p(K) = \left(\sum_{i \in K} |x_i|^p \right)^{\frac{1}{p}} \quad (17)$$

Temporal probability pooling is employed to refine sequential event detection by aggregating frame-level predictions over time. Following established temporal pooling strategies for sound event detection [34], the event-level probability is computed as:

$$y(e) = \frac{\sum_{t=1}^T y_t(e)^2}{\sum_{t=1}^T y_t(e)} \quad (18)$$

This formulation allows the model to identify both isolated and continuous acoustic threat events, producing context-aware threat confidence scores for further fusion. Fig. 3 depicts the overall procedure.

3.2.2. Visual threat analysis

The $Visual_{Deep}$ model integrates a CNN with an XGBoost classifier (ConvXGB) to analyze UAV-derived tactical imagery and sensor feeds. The CNN is employed for hierarchical feature extraction, while XGBoost performs the final classification based on learned representations. Following hybrid deep feature learning and gradient boosting frameworks [35], the training process can be expressed as:

$$ConvXGB = f_c(\phi(X_d^{train})) \quad (19)$$

where $\phi(\cdot)$ denotes the CNN-based feature extraction function and $f_c(\cdot)$ represents the XGBoost classifier. For unseen visual data, the inference process is defined as:

$$Y_c^j = f_c(\phi(X_d^{test})_j) \quad (20)$$

The CNN backbone employs convolutional layers with 20 and 15 filters, max pooling, dropout regularization (rate = 0.19), and dense layers with ReLU activations followed by a sigmoid output. The model is optimized using Adam with binary cross-entropy loss, balancing detection precision and real-time. Together, $Acoustic_{Deep}$ and $Visual_{Deep}$

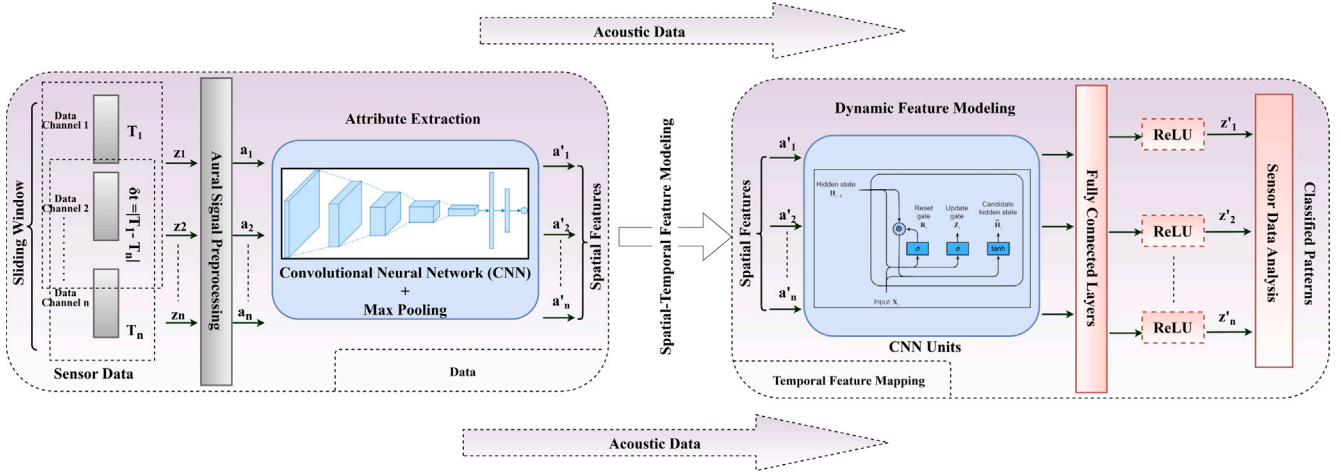


Fig. 3. Acoustic threat.

enable a comprehensive multimodal threat perception system. The fusion of acoustic and visual intelligence supports proactive detection of adversarial activities, weapon engagement cues, and movement anomalies—providing a foundation for the XAI-driven interpretation and decision layers in subsequent modules. Fig. 4 depicts the overall procedure.

3.3. Dynamic integration mechanism

The proposed framework employs a Dynamic Integration Mechanism (DIM) that extends the late-fusion paradigm through an adaptive Decision Tree (DT)-based learning process. Unlike traditional fusion strategies relying on fixed weighting or static aggregation rules, the DIM introduces a feedback-driven refinement cycle in which only misclassified threat instances are selectively retrained. This self-correcting capability enhances classification precision, mitigates overfitting, and sustains computational efficiency—representing a significant methodological advancement for real-time battlefield intelligence. The adaptability of the DIM ensures robust operation under heterogeneous, uncertain, or partially missing sensory inputs. The DT classifier was chosen as the foundation due to its inherent interpretability and resilience in handling incomplete multimodal data, allowing the system to maintain situational awareness even when specific sensory channels are unavailable. By dynamically retraining only on misclassified threat events, the DIM minimizes error propagation and supports continual learning with minimal processing overhead. Consequently, this mechanism yields improved accuracy in multimodal threat recognition and ensures consistent mission assurance decisions under variable combat conditions. The operational flow of the DIM is outlined as follows:

1. **Data Preprocessing:** Handle missing data points, normalize heterogeneous features, and encode categorical variables into machine-readable formats.
2. **Initial DT Model:** Train a baseline DT classifier on preprocessed multimodal data to establish primary decision boundaries.
3. **Error Detection:** Identify samples misclassified by the DT model through performance evaluation.
4. **Dynamic Retraining:** Re-optimize only the integration component using misclassified instances to refine decision boundaries while maintaining low computational cost.
5. **Threat Classification:** Generate final refined predictions using the dynamically integrated model, yielding enhanced detection of tactical and environmental threats.

This design capitalizes on the interpretability and efficiency of DTs while surpassing static late-fusion techniques by embedding a self-adaptive correction layer that sustains reliability and operational responsiveness. The complete integration workflow is summarized in Algorithm 1.

Algorithm 1 DIM for Multimodal Threat Recognition

Require: Multimodal dataset X , threat labels y

Ensure: Final threat prediction Y

```

1: procedure DIM( $X, y$ )
2:   Preprocess Data:
3:   Handle missing or corrupted values
4:   Normalize and standardize features
5:   Encode categorical variables numerically
6:   Train Base Model:
7:    $DT\_model \leftarrow \text{DECISIONTREECLASSIFIER}()$ 
8:    $DT\_model.fit(X, y)$ 
9:   Identify Misclassifications:
10:   $y\_pred \leftarrow DT\_model.predict(X)$ 
11:   $M \leftarrow \{x_i \in X \mid y_i \neq y\_pred[i]\}$ 
12:  Retrain Integration Layer:
13:   $DIM\_model \leftarrow \text{DYNAMICINTEGRATIONMODEL}()$ 
14:   $DIM\_model.fit(M, y[y \neq y\_pred])$ 
15:  Generate Final Prediction:
16:   $Y \leftarrow DIM\_model.predict(X)$ 
17:  return  $Y$ 
18: end procedure

```

3.4. Explainable-AI-driven prediction interpretation layer

Following the classification phase, the output predictions are processed through an XAI layer designed to provide mathematically grounded interpretability. The SHapley Additive exPlanations (SHAP) framework is utilized to quantify the marginal contribution of each input feature to the XGBoost model's output. Let the model output for an instance x be denoted as $f(x)$. The SHAP value for feature i , denoted by ϕ_i , is computed based on the Shapley value formulation [36] as:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F|-|S|-1)!}{|F|!} [f(S \cup \{i\}) - f(S)], \quad (21)$$

where F is the full set of features, S denotes a subset of features excluding feature i , and the weighting term $\frac{|S|!(|F|-|S|-1)!}{|F|!}$ represents

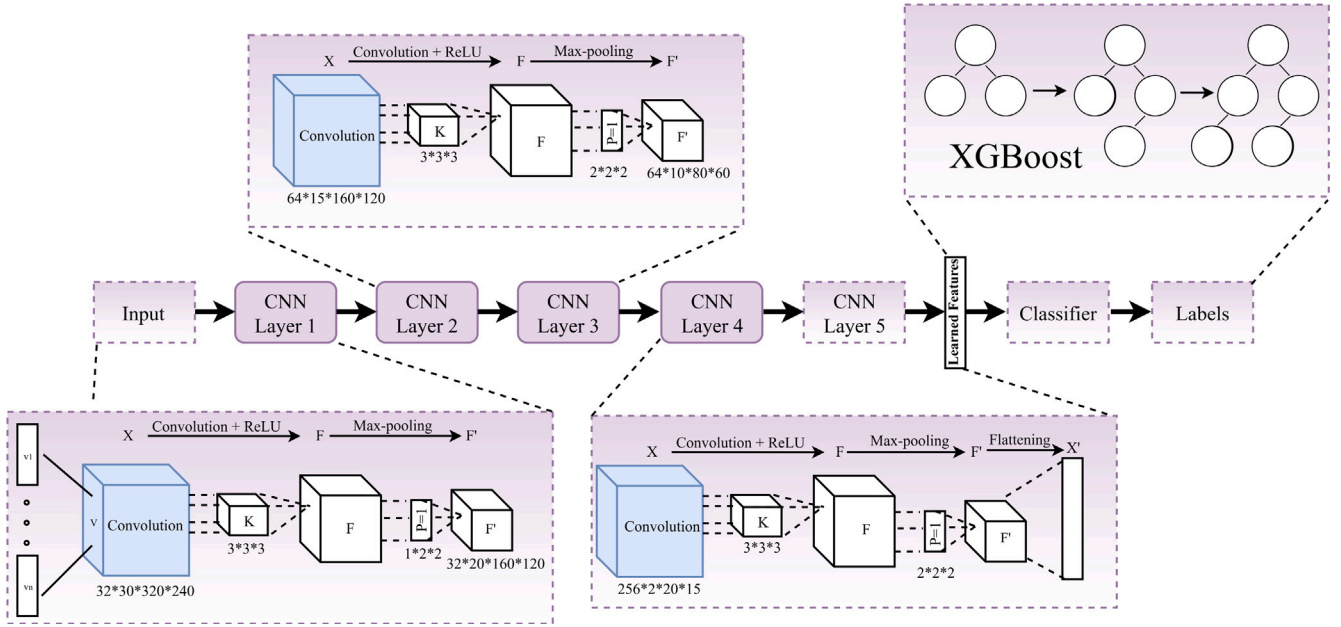


Fig. 4. Visual threat analysis.

the contribution of feature i across all possible coalitions. To enable practical computation, SHAP approximates this formulation using an expectation over a background dataset X_b , capturing the distribution of missing features. Accordingly, the final model prediction can be expressed as:

$$f(x) = \phi_0 + \sum_{i=1}^n \phi_i, \quad (22)$$

where $\phi_0 = \mathbb{E}_{x \sim X_b}[f(x)]$ denotes the expected model output over the background distribution. To ensure robustness and fairness, the SHAP framework is configured using a stratified subset of the training data as the background distribution, maintaining parity between threat-positive and threat-negative cases to minimize sampling bias. For global interpretability, $\text{Kernel}_{\text{SHAP}}$ is applied to $N_g = 1000$ randomly selected samples with a fixed random seed for reproducibility. For local interpretability, SHAP values $\phi_i^{(x_j)}$ are computed per test instance x_j , yielding case-specific attributions that explain individual predictions. The computed SHAP values quantify how each multimodal feature affects the final threat classification, effectively bridging model logic and operational reasoning. Globally aggregated SHAP values are computed as [37];

$$\bar{\phi}_i = \frac{1}{N_g} \sum_{j=1}^{N_g} \phi_i^{(x_j)} \quad (23)$$

It reveal dominant threat indicators across the dataset, whereas local values highlight the decisive factors influencing a specific event or mission instance. Unlike traditional XAI techniques that provide static visualizations, the proposed SHAP-based layer emphasizes actionable interpretability. The feature attributions are dynamically mapped to operational decisions such as risk prioritization, mission alert verification, and adaptive resource deployment. This ensures transparency, supports human-in-the-loop decision-making, and enhances the overall reliability and accountability of the multimodal threat recognition framework. The complete interpretability workflow and its integration within the prediction pipeline are depicted in Algorithm 2.

Algorithm 2 Local-Global SHAP Computation for Multimodal Threat Interpretation

Require: Trained XGBoost model $f(\cdot)$, multimodal input data X , background dataset X_b

Ensure: Global and local SHAP attributions $\Phi_{\text{global}}, \Phi_{\text{local}}$

1: **Initialization:**

2: Define feature set $F = \{f_1, f_2, \dots, f_n\}$ and background distribution X_b

3: Initialize global attribution accumulator $\Phi_{\text{global}} = \mathbf{0}$

4: **For each test instance** $x_j \in X$:

5: Compute expected model output:

$$\phi_0 = \mathbb{E}_{x \in X_b}[f(x)]$$

6: **For each feature** $f_i \in F$:

7: Sample random subsets $S \subseteq F \setminus \{f_i\}$

8: Estimate marginal contribution:

$$\phi_i^{(x_j)} = \mathbb{E}_S[f(S \cup \{f_i\}) - f(S)]$$

9: Aggregate local contributions:

$$f(x_j) = \phi_0 + \sum_{i=1}^n \phi_i^{(x_j)}$$

10: Store $\Phi_{\text{local}}(x_j) = \{\phi_1^{(x_j)}, \dots, \phi_n^{(x_j)}\}$

11: **End for**

12: **Compute global importance:**

$$\Phi_{\text{global}}(f_i) = \frac{1}{|X|} \sum_{j=1}^{|X|} |\phi_i^{(x_j)}|$$

13: **return** $\Phi_{\text{global}}, \Phi_{\text{local}}$

4. Performance evaluation

This section presents a comprehensive performance evaluation of the proposed XAI-driven DTW-enabled battlefield intelligence framework for multimodal threat assessment. Both acoustic and visual data

streams are analyzed to evaluate the framework's operational accuracy, robustness, synchronization capability, and adaptive decision-support performance under dynamic battlefield conditions. The experiments are designed to validate the framework's dynamic integration mechanism, interpretability layer, and overall suitability for mission-critical decision support in defense environments. The evaluation comprises the following components:

1. Experimental configuration and setup;
2. Modality-specific threat analysis:
 - Acoustic anomaly detection and classification,
 - Visual threat identification via UAV imagery,
 - Dynamic integration of multimodal evidence for risk estimation;
3. Analysis of system reliability and temporal stability under variable battlefield conditions;
4. XAI-based interpretation and feature attribution.

4.1. Experimental configuration

The evaluation employs DL and XAI methodologies to model and interpret battlefield events. CNN-based feature extractors and XGBoost classifiers are integrated with SHAP-based explainability to quantify and visualize the relative impact of input features on threat predictions. This hybrid architecture enhances both diagnostic accuracy and operational transparency. The prediction function is defined as:

$$Y = f_{\theta}(X_A, X_V)$$

where X_A and X_V denote acoustic and visual modalities, respectively, and θ represents trainable parameters optimized via stochastic gradient descent. The SHAP attribution for each feature f_i is given by:

$$\phi_i = \mathbb{E}_{S \subseteq F \setminus \{f_i\}} [f(S \cup \{f_i\}) - f(S)]$$

which provides an interpretable contribution measure toward the final threat classification.

Dataset description and experimental design The experimental evaluation was conducted using a combination of publicly available benchmark datasets and synthetically generated mission-oriented multimodal data to ensure diversity, realism, robustness, and reproducibility in battlefield threat assessment scenarios. The experimental framework was designed to emulate dynamic battlefield environments involving heterogeneous sensory inputs, evolving operational conditions, and real-time cyber-physical interactions.

Acoustic Modality Dataset: For the acoustic modality, environmental sound datasets containing anomalous and hostile acoustic signatures were utilized, including the publicly available *ESC-50* dataset.¹ The dataset contains labeled environmental sound events spanning multiple categories relevant to anomaly detection and battlefield situational awareness.

Acoustic Preprocessing: The acoustic signals underwent a standardized preprocessing pipeline consisting of:

- resampling to 44.1 kHz,
- silence trimming and amplitude normalization,
- noise filtering using spectral gating,
- segmentation into fixed-duration overlapping windows,
- extraction of Mel-Frequency Cepstral Coefficients (MFCCs),
- temporal spectrogram generation for CNN-GRU learning.

Visual Modality Dataset: For the visual modality, UAV surveillance imagery and aerial monitoring datasets were employed, including the publicly available *VisDrone* dataset.² The dataset provides drone-captured scenes containing moving objects, urban activity, and dynamic operational environments suitable for surveillance-oriented threat assessment.

Visual Preprocessing: The visual preprocessing pipeline included:

- image resizing and spatial normalization,
- histogram equalization for illumination correction,
- frame stabilization and motion compensation,
- contrast enhancement,
- object-centric region extraction,
- data augmentation using rotation, flipping, scaling, blur simulation, and environmental perturbations.

Synthetic Battlefield Scenario Generation: To represent mission-critical battlefield conditions that cannot be publicly released due to operational confidentiality and security restrictions, the study additionally incorporated synthetically generated multimodal defense scenarios. The synthetic data generation framework was designed to emulate realistic battlefield interactions while maintaining transparency and reproducibility.

Dynamic Battlefield Simulation Parameters: The synthetic battlefield environment was generated using configurable simulation parameters, including:

- troop and hostile object density,
- UAV flight trajectories and altitude variation,
- target motion speed and direction,
- environmental noise intensity,
- acoustic interference levels,
- visibility degradation due to smoke, fog, or camouflage,
- sensor latency and communication delay,
- threat severity and adversarial activity levels.

Synthetic Data Generation Process: The synthetic multimodal data generation pipeline consisted of the following stages:

1. **Scenario Modeling:** Parameterized battlefield templates were created to simulate hostile intrusion, abnormal movement, UAV surveillance events, tactical vehicle motion, acoustic disturbances, environmental anomalies, and adversarial concealment conditions.
2. **Visual Data Synthesis:** Public UAV imagery from the *VisDrone* dataset was augmented using geometric transformations, motion blur simulation, illumination variation, fog and smoke injection, object occlusion, camouflage perturbation, and adversarial environmental effects. Synthetic hostile entities and abnormal trajectories were inserted to emulate realistic battlefield threats.
3. **Acoustic Signal Generation:** ESC-50 audio samples were combined with synthetic battlefield sounds, including drone noise, explosion-like events, gunfire impulses, vehicle movement, mechanical anomalies, and environmental disturbances. Reverberation modeling, Gaussian noise injection, temporal shifting, and amplitude modulation were applied to improve realism.
4. **Multimodal Synchronization Mechanism:** Visual and acoustic streams were temporally aligned using timestamp-based synchronization and event-correlation mapping. Each synchronized event pair represented a unified battlefield activity instance, enabling cross-modal feature learning and multimodal threat correlation analysis.
5. **Threat Label Generation:** Threat labels were generated using rule-based operational criteria derived from predefined battlefield conditions. Labels included:

¹ Source: <https://github.com/karolpiczak/ESC-50>.

² Source: <https://github.com/VisDrone/VisDrone-Dataset>.

- normal operational state,
- suspicious activity,
- hostile intrusion,
- abnormal acoustic anomaly,
- high-risk tactical threat.

Label assignment considered multimodal consistency, threat intensity, environmental context, and adversarial event occurrence.

- Data Augmentation and Balancing:** To improve robustness and reduce class imbalance, additional augmentation strategies, including random cropping, spectral masking, temporal scaling, signal perturbation, Gaussian noise injection, and adversarial augmentation, were applied across both modalities.

Dataset Partitioning and Cross-Validation Strategy: To ensure unbiased evaluation and eliminate information leakage, a two-stage dataset partitioning strategy was adopted. Initially, the complete dataset was divided into:

- **Development Set (80%):** Used for training, hyperparameter optimization, and cross-validation.
- **Test Set (20%):** Held out entirely and used only for final evaluation.

Subsequently, the development set was processed using stratified 5-fold cross-validation consistently across both acoustic and visual modalities:

- four folds were used for training,
- one fold was used for validation,
- the process was repeated across all folds,
- average validation performance was computed for model selection.

The validation partitions were exclusively used for:

- hyperparameter tuning,
- early stopping,
- model selection,
- optimization refinement.

The independent test set remained completely unseen throughout training and optimization to ensure unbiased generalization assessment.

Reproducibility Measures: To facilitate independent validation and replication:

- standardized preprocessing pipelines were applied,
- fixed random seeds were maintained,
- hyperparameter settings were documented,
- synchronization procedures were consistently enforced across modalities,
- experimental configurations and augmentation procedures were systematically recorded.

Although certain operationally sensitive datasets cannot be publicly disclosed, the detailed synthetic data generation methodology, multimodal synchronization framework, preprocessing procedures, and use of publicly accessible datasets provide sufficient transparency for reproducibility and future comparative studies.

Structured hyperparameter optimization framework: The optimal hyperparameters reported in Table 3 were determined using a structured multi-stage optimization framework designed to balance predictive performance, generalization capability, and computational efficiency across multimodal battlefield threat assessment tasks.

Stage 1: Definition of Bounded Search Space Initially, bounded search spaces were systematically defined for each hyperparameter based on:

- prior empirical findings from multimodal deep learning literature,
- architectural constraints of the CNN-GRU and CNN-based models,

- computational resource limitations,
- convergence stability considerations.

Lower and upper bounds were selected to avoid unstable optimization behavior, excessive computational overhead, and model overparameterization. For example, smaller learning rates improved convergence stability but increased training time, whereas excessively large learning rates caused unstable gradient updates. Similarly, deeper CNN architectures enhanced feature representation capability but substantially increased latency and overfitting risk.

Stage 2: Random Search-Based Coarse Exploration A random search strategy was initially employed for coarse-grained exploration of the high-dimensional hyperparameter space. Compared to exhaustive grid search, random search provides better efficiency in identifying high-performing parameter regions while significantly reducing computational complexity. During this stage:

- random parameter combinations were sampled from the predefined bounded search space,
- each configuration was evaluated using validation performance,
- poorly performing regions were discarded early to reduce optimization cost.

Stage 3: Bayesian Optimization-Based Fine Tuning After identifying promising parameter regions, Bayesian optimization was applied for fine-grained hyperparameter refinement. Bayesian optimization modeled the validation objective as a probabilistic surrogate function and iteratively selected candidate configurations that maximized expected performance improvement. The optimization objective was defined as:

$$\max_{\theta} J(\theta) = F1_{val} - \lambda \cdot Loss_{val} \quad (24)$$

where:

- θ denotes the hyperparameter configuration,
- $F1_{val}$ represents validation F1-score,
- $Loss_{val}$ denotes validation loss,
- λ controls the trade-off between predictive performance and optimization stability.

All optimization decisions were performed exclusively using development-set validation performance to prevent information leakage.

Stage 4: 5-Fold Cross-Validation and Early Stopping To improve robustness and generalization reliability, 5-fold stratified cross-validation was incorporated during hyperparameter tuning. The development set (80%) was partitioned into five stratified folds while preserving class distribution consistency across acoustic and visual modalities. For each fold:

- four folds were used for training,
- one fold was used for validation,
- the process was repeated five times,
- average validation performance was computed across folds.

Additionally, early stopping with a patience value of 10 epochs was employed based on validation loss minimization:

$$\text{Stop Training if } Loss_{val}^{(t)} > Loss_{val}^{(t-p)} \quad (25)$$

where $p = 10$ denotes the patience parameter. This mechanism prevented overfitting, reduced unnecessary training iterations, and improved computational efficiency.

Influence of Hyperparameters on Model Performance: The selected hyperparameters significantly influenced prediction accuracy, generalization capability, and computational cost:

- **Learning Rate:** Controlled optimization stability and convergence speed.

Table 3

Optimized hyperparameter configuration using multi-stage hybrid optimization framework (Development set: 80%, internally partitioned using 5-fold cross-validation; Final test set: 20%).

Model	Hyperparameter	Search range	Optimized value
Proposed	Learning Rate	$\{1e^{-5}, 5e^{-4}, 1e^{-3}, 1e^{-2}\}$	$1e^{-3}$
	Training epochs	$\{30, 50, 80, 100, 120\}$	100
	Batch size	$\{16, 32, 64, 128\}$	32
	Audio sampling rate	$\{22,050 \text{ Hz}, 44,100 \text{ Hz}, 48,000 \text{ Hz}\}$	44,100 Hz
	CNN layers	$\{3, 5, 7, 9, 11\}$	7
	MFCC coefficients	$\{13, 20, 26, 40\}$	40
	Activation function	$\{\text{ReLU}, \text{LeakyReLU}, \text{GELU}\}$	GELU
	GRU units	$\{64, 128, 256\}$	128
	Dropout rate	$\{0.1, 0.2, 0.3, 0.4\}$	0.3
	Optimizer	$\{\text{Adam}, \text{AdamW}, \text{RMSProp}\}$	AdamW
	Loss function	$\{\text{Cross-entropy}, \text{Sparse categorical CE}, \text{Focal loss}\}$	Sparse categorical cross-entropy
	Development: Test split	$\{70:30, 80:20, 85:15\}$	80:20

- **CNN Depth:** Increased feature extraction capability, but also raised computational latency and overfitting risk.
- **GRU Units:** Improved temporal dependency learning for acoustic sequences while increasing memory consumption.
- **Dropout Rate:** Reduced overfitting and improved generalization across multimodal scenarios.
- **Batch Size:** Influenced gradient stability, GPU utilization, and training throughput.
- **MFCC Coefficients:** Controlled the richness of acoustic spectral representation.
- **Optimizer Selection:** AdamW improved convergence stability and regularization performance compared to standard optimizers.

Stage 5: Final Model Selection and Evaluation The final hyperparameter configuration was selected using a multi-objective optimization criterion balancing predictive performance and inference efficiency:

$$\theta^* = \arg \max_{\theta} (\alpha \cdot F1_{val} + \beta \cdot \text{Accuracy}_{val} - \gamma \cdot \text{Latency}) \quad (26)$$

After hyperparameter selection, the finalized models were retrained using the combined training and validation partitions of the development set and evaluated once on the held-out test set (20%). All experiments were conducted using fixed random seeds, standardized preprocessing pipelines, and consistent dataset partitions to ensure reproducibility, transparency, and unbiased generalization assessment.

Evaluation metrics: To ensure robust and reproducible evaluation, model performance on the test set was assessed using bootstrapped sampling to compute 95% confidence intervals (CIs). The evaluation is based on standard classification metrics derived from the confusion matrix, including true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). The following quantitative metrics were used:

1. **Recall (REC):** Measures the ability of the model to correctly identify true threat instances:

$$\text{REC} = \frac{TP}{TP + FN} \quad (27)$$

2. **Precision (PRE):** Evaluates the proportion of correctly predicted threat instances among all predicted positives, reflecting the model's ability to minimize false alarms:

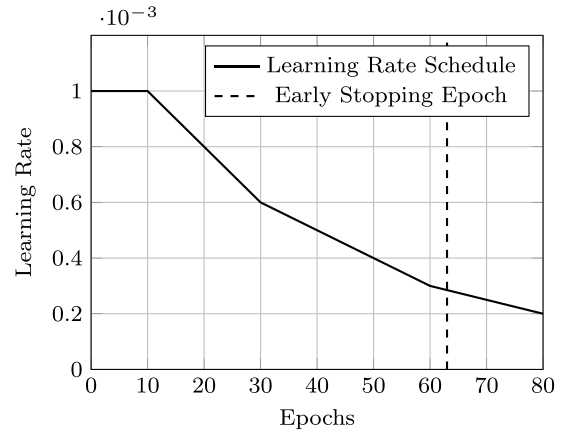
$$\text{PRE} = \frac{TP}{TP + FP} \quad (28)$$

3. **Accuracy (ACC):** Represents the overall correctness of the model across all classes:

$$\text{ACC} = \frac{TP + TN}{TP + TN + FP + FN} \quad (29)$$

4. **F1-Score:** Provides a balanced measure of precision and recall, particularly useful in imbalanced datasets:

$$F1 = 2 \cdot \frac{\text{PRE} \cdot \text{REC}}{\text{PRE} + \text{REC}} \quad (30)$$

**Fig. 5.** Learning rate schedule with early stopping criterion.**Table 4**

Training and validation accuracy over epochs.

Number of epochs	Training accuracy (%)	Validation accuracy (%)
5	78	80
10	82	83
15	84	85
20	86	87
25	89	90
30	92	93
35	94	94
40	95	95
45	96	95
50	96	96
55	97	96
60	97	97
65	97	97
70	98	97
75	98	98
80	98	98

5. **Specificity (SPEC):** Measures the ability of the model to correctly identify non-threat instances:

$$\text{SPEC} = \frac{TN}{TN + FP} \quad (31)$$

These metrics collectively provide a comprehensive evaluation of the model's detection capability, balancing sensitivity to threats and robustness against false positives, which is critical for mission-critical defense applications.

4.2. Evaluation of threat signatures

The proposed framework was evaluated using a stratified 5-fold cross-validation protocol applied exclusively on the development set

Table 5

Training and validation loss over epochs.

Number of epochs	Training loss (%)	Validation loss (%)
5	2.8	2.5
10	2.4	2.2
15	2.0	1.9
20	1.8	1.6
25	1.5	1.4
30	1.3	1.2
35	1.1	1.0
40	0.9	0.8
45	0.8	0.7
50	0.7	0.6
55	0.6	0.5
60	0.5	0.4
65	0.4	0.3
70	0.3	0.25
75	0.25	0.2
80	0.2	0.15

(80%) to ensure reliability and generalization across multimodal inputs. This design is consistent with the dataset partitioning strategy, where the test set (20%) remains completely unseen during model training and validation. The development set was partitioned into five folds with preserved class distribution. In each iteration, four folds (i.e., 80% of the development set) were used for training, while the remaining fold (20%) served as the validation set. A fixed random seed (40) was used to ensure reproducibility. The aggregate classification accuracy across folds is defined as:

$$ACC = \frac{1}{k} \sum_{i=1}^k ACC_i, \quad k = 5 \quad (32)$$

4.2.1. Acoustic pattern analysis

For the acoustic modality, auditory streams were processed using Mel-Frequency Cepstral Coefficients (MFCC) with adaptive sampling and temporal segmentation. Data augmentation techniques, including time-shifting, spectral masking, and Gaussian noise injection, were applied to enhance model generalization under varying environmental conditions. The acoustic model, based on a CNN-GRU architecture, was trained using the AdamW optimizer [38] for a maximum of 100 epochs with early stopping (patience = 10) based on validation loss. The final hyperparameter configuration corresponds to the optimal settings obtained through the proposed hybrid optimization strategy, ensuring a balance between detection accuracy, robustness, and computational efficiency.

4.2.2. Explainable threat detection model evaluation and performance analysis

The proposed model is designed to enhance interpretability and robustness in battlefield threat pattern recognition within a DTW-enabled defense environment. It integrates deep convolutional and recurrent layers to capture nonlinear spatiotemporal representations from multimodal battlefield data, including acoustic signals and UAV-based visual streams. Convolutional layers with ReLU activation functions, followed by max-pooling operations, are employed to extract hierarchical features while suppressing redundant noise from sensor inputs. The recurrent (GRU-based) component captures temporal dependencies, particularly in the acoustic modality, enabling effective modeling of sequential threat patterns. To ensure stable and efficient convergence, multiple learning rate schedules were explored, balancing rapid initial learning with gradual refinement. An early stopping mechanism (patience = 10 epochs) was implemented based on validation loss to prevent overfitting, with the best-performing model weights restored using Keras/TensorFlow callbacks. Model stability was achieved with a batch size of 32, with consistent generalization observed up to approximately 63 epochs, beyond which minor overfitting began to emerge. The learning rate schedule (Fig. 5) provides

insight into convergence behavior and optimization dynamics. Table 4 presents the progression of training and validation accuracy, while Table 5 illustrates the corresponding loss trends. The model achieved high validation performance, indicating stable convergence and reproducibility under controlled experimental conditions with fixed random seed initialization.

Threat Classification Performance: Evaluation was conducted using stratified 5-fold cross-validation applied to the development set, ensuring consistency with the predefined dataset partitioning strategy. In each fold, the model was trained and validated on distinct subsets while preserving class distribution, enabling robust assessment of generalization across diverse battlefield conditions. Performance metrics were computed across all folds. The confusion matrix was used to quantify the model's threat discrimination capability. As shown in Tables 6 and 7, the proposed XAI-DT framework demonstrates consistent and interpretable classification performance across multiple battlefield threat modalities. Minor variations are observed in scenarios involving overlapping or concurrent threats; however, the model maintains stable convergence and reliable predictions. The integration of explainability mechanisms, including layer-wise relevance propagation and SHAP-based feature attribution, further enhances transparency by providing insight into the decision-making process. The model achieves an average accuracy of 77.30% and precision of 77.69%, supporting its suitability for real-time, explainable threat assessment within DTW-enabled operational environments.

4.2.3. Comparative analysis

Methodology: To evaluate the effectiveness and interpretability of the proposed framework, a comprehensive comparative study was conducted against established deep learning architectures, including CNN, RNN, and LSTM. In addition, the proposed model was qualitatively compared with recent state-of-the-art (SOTA) approaches reported in the literature, including: (i) data-driven multimodal threat detection frameworks based on deep neural networks [39], (ii) adaptive anomaly detection models using temporal sequence learning [40], and (iii) hybrid AI-based situational awareness systems integrating heterogeneous sensor data [41]. These approaches represent recent advancements in intelligent threat assessment and serve as strong baselines for benchmarking. All models were trained and optimized using the development set (80%), ensuring consistency with the predefined dataset partitioning strategy. The final evaluation was performed exclusively on the held-out test set (20%), which remained completely unseen during training and model selection. This protocol ensures an unbiased assessment of generalization performance.

Evaluation Protocol: The results reported in Tables 8 and 9 correspond solely to test set performance. Standard classification metrics, including accuracy, precision, recall, F1-score, and specificity, are reported along with 95% CIs computed via bootstrapping to ensure statistical robustness and reproducibility.

Results and statistical analysis: The results presented in Table 8 demonstrate that the proposed XAI-DT framework consistently outperforms conventional deep learning baselines across all evaluated battlefield threat scenarios on the unseen test set (20%). The inclusion of 95% confidence intervals further validates the statistical reliability, robustness, and generalization capability of the proposed framework under dynamic operational conditions. For hostile signal detection, the proposed framework achieved the highest classification accuracy of $97.06\% \pm 0.84$, along with strong precision, recall, and specificity values. The relatively narrow confidence intervals indicate stable model behavior and reduced prediction variance across multiple validation folds. Compared to CNN-, RNN-, and LSTM-based approaches, the proposed framework demonstrated superior sensitivity to hostile signal patterns while minimizing both false positives and missed threat events. In anomalous behavior recognition, conventional baseline models exhibited high precision but comparatively lower recall values, indicating limited sensitivity toward subtle or evolving anomalous activities. In contrast, the

Table 6
5-fold validation performance (F_1 – F_5) across multimodal battlefield threat indicators.

Fold	Threat type	Projectile launch	Vehicle intrusion	Aerial recon activity	Explosive detonation
F_1	Projectile launch	77.59%	0.09%	0.62%	0.12%
	Vehicle intrusion	0.14%	78.88%	0.00%	0.04%
	Aerial recon activity	0.37%	0.00%	76.06%	0.23%
	Explosive detonation	0.13%	0.00%	0.11%	78.64%
F_2	Projectile launch	76.72%	0.19%	0.69%	0.25%
	Vehicle intrusion	0.16%	79.00%	0.00%	0.12%
	Aerial recon activity	0.03%	0.00%	77.77%	0.09%
	Explosive detonation	0.08%	0.02%	0.13%	79.00%
F_3	Projectile launch	78.45%	0.03%	0.23%	0.15%
	Vehicle intrusion	0.21%	78.49%	0.55%	0.02%
	Aerial recon activity	0.14%	0.34%	76.93%	0.58%
	Explosive detonation	0.06%	0.28%	0.33%	78.81%
F_4	Projectile launch	77.51%	0.24%	0.48%	0.06%
	Vehicle intrusion	0.33%	79.00%	0.00%	0.53%
	Aerial recon activity	0.17%	0.00%	77.71%	0.34%
	Explosive detonation	0.04%	0.00%	0.27%	79.00%
F_5	Projectile launch	79.00%	0.50%	0.09%	0.27%
	Vehicle intrusion	0.39%	78.93%	0.27%	0.21%
	Aerial recon activity	0.17%	0.00%	77.71%	0.34%
	Explosive detonation	0.04%	0.00%	0.27%	79.00%

Table 7
Aggregated performance metrics across threat detection categories.

Threat type	ACC	PRE	REC	F1	SPEC
Projectile launch	77.40%	77.51%	75.87%	77.51%	77.48%
Vehicle intrusion	77.48%	78.89%	74.35%	73.61%	77.02%
Aerial recon activity	76.53%	75.64%	74.19%	74.48%	75.26%
Explosive detonation	77.85%	79.00%	74.54%	78.06%	78.36%
Mean	77.30%	77.69%	74.50%	75.68%	77.00%

proposed XAI-DT framework achieved a more balanced performance profile with an accuracy of $97.17\% \pm 0.79$, improving anomaly detection sensitivity while maintaining high precision and stable generalization performance. The reduced variability reflected by the confidence intervals suggests improved robustness against heterogeneous behavioral patterns and operational uncertainty. For sensor data anomaly detection, the proposed framework attained the highest overall performance with an accuracy of $97.64\% \pm 0.72$ and specificity of $98.39\% \pm 0.58$. These results demonstrate the effectiveness of the multimodal DTW architecture in modeling complex spatiotemporal battlefield dependencies and distinguishing anomalous sensor activity from normal operational behavior. The low confidence interval range further indicates strong consistency and resilience to noisy sensory environments. In communication interference scenarios, the proposed model maintained stable and reliable performance with an F1-score of $93.49\% \pm 1.13$. Despite the presence of dynamic environmental interference, signal degradation, and operational uncertainty, the framework exhibited improved robustness and adaptive generalization compared to traditional deep learning baselines. Overall, the comparative analysis indicates that the proposed XAI-DT framework not only improves predictive accuracy but also achieves stronger statistical stability, reduced performance variance, and enhanced operational reliability across diverse threat conditions. The integration of multimodal learning, Digital Twin synchronization, and explainable AI mechanisms contributes significantly to robust real-time battlefield threat assessment, thereby supporting deployment in mission-critical defense environments.

Discussion A comparison with recent state-of-the-art (SOTA) studies further highlights the effectiveness of the proposed framework. Data-driven multimodal models such as Latif et al. [39] demonstrate strong predictive performance; however, they are typically constrained by static feature fusion strategies and limited interpretability. As a result, their decision-making process remains opaque, which restricts their applicability in mission-critical environments. Similarly, temporal

sequence-based anomaly detection approaches [40] effectively capture sequential dependencies but often lack robust multimodal integration and contextual awareness, leading to reduced performance in complex, dynamic scenarios. Hybrid AI frameworks [41] improve situational awareness by combining heterogeneous data sources; however, they predominantly rely on black-box models with limited explainability and reduced transparency in decision-making. In contrast, the proposed XAI-DT framework integrates acoustic and visual modalities within a DT environment, enabling dynamic, context-aware learning and adaptive threat assessment. The incorporation of explainable AI techniques, including SHAP-based feature attribution and LRP, provides both global and instance-level interpretability, thereby enhancing transparency and supporting human-in-the-loop decision-making. The superior performance observed in the experimental results — particularly in terms of accuracy, recall, and specificity — can be attributed to (i) effective multimodal fusion, (ii) robust spatiotemporal feature extraction, and (iii) the integration of explainability mechanisms that improve model reliability under uncertain conditions. Overall, the proposed framework advances the state-of-the-art by not only improving predictive performance but also addressing critical limitations related to interpretability, adaptability, and robustness. These characteristics make it well-suited for real-time deployment in mission-critical defense applications, where both accuracy and transparency are essential.

4.2.4. Analysis of patterns based on visual intelligence samples

The proposed framework leverages the PyTorch platform integrated with the Fastai library for efficient development, training, and deployment of the visual intelligence module. The evaluation utilizes UAV-based and aerial imagery datasets, including publicly available benchmarks and curated domain-specific samples, as described in the dataset section. Consistent with the overall experimental design, the visual modality is trained using the development set (80%) with stratified sampling, while final evaluation is conducted on the held-out test set (20%). Within the development set, model validation is performed using stratified cross-validation to ensure robustness and generalization across diverse operational scenarios. To enhance convergence stability and training efficiency, a controlled learning rate schedule was employed in conjunction with an early stopping mechanism (patience = 10 epochs). The learning rate was progressively reduced during training to prevent oscillations and enable fine-grained optimization near convergence. Early stopping was triggered when the validation loss plateaued, ensuring that overfitting was minimized while preserving the best-performing model weights. Model optimization was performed using the Adam optimizer with an initial learning rate of 0.001 for a

Table 8

Comparative analysis of threat pattern recognition models on the held-out test set (20%) with 95% confidence intervals.

Threat scenario	Model	ACC (%)	PRE (%)	REC (%)	F1-Score (%)	SPEC (%)
Hostile signal detection	Proposed (XAI-DT)	97.06 ± 0.84	97.33 ± 0.79	95.25 ± 1.12	97.33 ± 0.81	97.29 ± 0.76
	CNN	91.58 ± 1.34	85.66 ± 1.72	87.33 ± 1.58	86.49 ± 1.63	94.21 ± 1.11
	RNN	90.85 ± 1.46	90.11 ± 1.39	81.26 ± 1.94	85.46 ± 1.68	89.71 ± 1.43
	LSTM	93.27 ± 1.18	91.73 ± 1.29	90.62 ± 1.34	89.69 ± 1.42	92.96 ± 1.17
Anomalous behavior recognition	Proposed (XAI-DT)	97.17 ± 0.79	99.07 ± 0.52	93.32 ± 1.21	92.39 ± 1.16	96.71 ± 0.88
	CNN	93.65 ± 1.16	99.21 ± 0.48	87.75 ± 1.42	89.64 ± 1.35	95.21 ± 1.03
	RNN	92.92 ± 1.23	99.21 ± 0.47	84.86 ± 1.61	86.63 ± 1.49	94.71 ± 1.12
	LSTM	91.99 ± 1.31	99.21 ± 0.45	88.95 ± 1.33	88.73 ± 1.28	94.96 ± 1.06
Sensor data anomaly	Proposed (XAI-DT)	97.64 ± 0.72	99.21 ± 0.41	93.56 ± 1.08	98.02 ± 0.63	98.39 ± 0.58
	CNN	93.35 ± 1.19	94.86 ± 1.08	94.07 ± 1.13	92.46 ± 1.27	95.71 ± 0.97
	RNN	91.74 ± 1.34	92.65 ± 1.21	92.09 ± 1.25	90.79 ± 1.38	94.21 ± 1.11
	LSTM	92.57 ± 1.28	94.05 ± 1.14	92.19 ± 1.17	94.64 ± 1.03	95.46 ± 0.96
Communication interference	Proposed (XAI-DT)	95.93 ± 0.96	94.95 ± 1.02	93.12 ± 1.19	93.49 ± 1.13	94.47 ± 0.98
	CNN	92.88 ± 1.17	93.77 ± 1.09	92.89 ± 1.11	91.73 ± 1.24	92.38 ± 1.15
	RNN	90.78 ± 1.36	91.62 ± 1.24	90.83 ± 1.29	90.06 ± 1.31	93.71 ± 1.07
	LSTM	91.66 ± 1.29	92.69 ± 1.18	91.10 ± 1.21	89.55 ± 1.37	92.96 ± 1.14

Table 9

Quantitative and qualitative comparison of the proposed framework with recent state-of-the-art approaches.

Method	Accuracy (%)	F1-Score (%)	Data modality	Key strength	Explainability
Latif et al. [39]	88–92	85–90	Multimodal	Effective deep feature extraction	No
Darban et al. [40]	87–91	84–89	Temporal (Sequential)	Strong temporal dependency modeling	No
Alshuhail et al. [41]	89–93	86–91	Heterogeneous sensors	Hybrid learning for situational awareness	Limited
Proposed XAI-DT framework	95.93–97.64	92.39–98.02	Multimodal (Acoustic + Visual)	Context-aware DTW integration	Yes

Table 10

Optimal hyperparameter configurations for the proposed XAI-DT framework.

Model	Hyper-parameter	Range	Optimized value
Proposed (XAI-DT)	Learning rate	{0.0003, 0.001, 0.01}	0.001
	Training epochs	{20, 40, 60, 80, 100}	50
	Batch size	{16, 32, 64}	16
	Convolution layers	{11, 12, 13}	11
	Activation function	{ReLU, LeakyReLU, GELU}	ReLU
	Optimizer	{Adam, AdamW, RMSProp}	Adam
	Data split ratio	{70:30, 80:20, 85:15}	80% Training & 20% Testing
	Loss function	{Cross entropy, Focal loss}	Cross entropy

maximum of 100 epochs. However, based on validation performance, optimal convergence was achieved at approximately 50 epochs, as illustrated in Fig. 6. This training strategy enables the visual model to effectively capture hierarchical spatial features and object-level representations from complex reconnaissance imagery, supporting reliable and robust threat pattern recognition under varying environmental conditions.

Hyperparameter selection and optimization: The optimal hyperparameters presented in Table 10 were obtained through a systematic and data-driven optimization process. Initially, a constrained search space was defined for each parameter based on prior empirical studies and architectural considerations of the proposed XAI-DT framework. To efficiently explore this space, a hybrid optimization strategy was adopted. Random search was first employed for coarse-grained exploration across candidate configurations, enabling the identification of promising regions without exhaustive enumeration. Subsequently, Bayesian optimization was applied to refine these configurations by modeling validation performance as a probabilistic objective function, thereby guiding the search toward optimal parameter combinations. Model performance during tuning was evaluated using stratified 5-fold cross-validation on the development set to ensure robustness across varying data distributions. Early stopping (patience = 10 epochs) was incorporated to prevent overfitting and automatically determine the optimal number of training epochs by monitoring validation loss. The model parameters corresponding to the lowest validation loss were retained. The final hyperparameter set was selected based on validation performance, prioritizing F1-score and accuracy while maintaining computational efficiency. Additionally, a fixed random seed

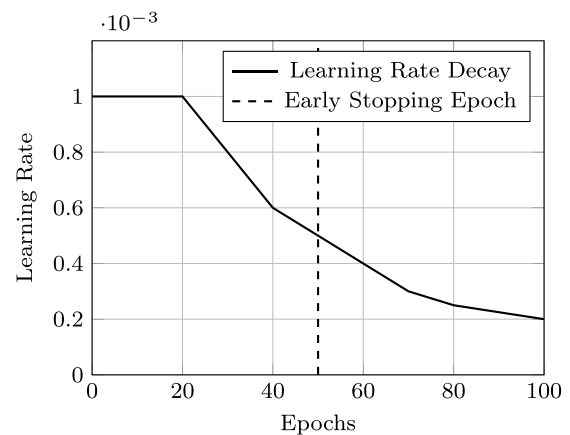


Fig. 6. Learning rate schedule with early stopping. The model achieves optimal convergence at epoch 50, after which validation performance stabilizes and early stopping is triggered.

(42) and consistent data splits were used to ensure reproducibility. To enhance robustness and generalization in the visual modality, data augmentation techniques such as controlled rotation, scaling, and horizontal flipping were applied. Augmentation was carefully constrained within realistic bounds to avoid introducing distortions inconsistent with operational imagery, thereby preserving the spatial and semantic

Table 11
Training and validation accuracy.

Number of epochs	Accuracy (%)
5	95.35
10	95.98
15	96.37
20	96.83
25	97.14
30	97.59
35	98.12
40	98.07
45	98.27
50	98.84

Table 12
Training and validation loss with increasing number of epochs.

Number of epochs	Loss
5	0.078
10	0.065
15	0.052
20	0.047
25	0.038
30	0.025
35	0.018
40	0.011
45	0.005
50	0.003

integrity of threat patterns. This multi-stage optimization framework ensures that the selected hyperparameters achieve a balanced trade-off between predictive accuracy, generalization capability, and real-time deployment feasibility. The test set was subsequently employed to assess the model's interpretability and accuracy in recognizing tactical threat patterns and situational anomalies. Tables 11 and 12 present the model's training accuracy and loss trends, respectively. Table 13 presents the performance comparison of the proposed XAI-DT model with state-of-the-art frameworks on visual intelligence datasets. To further contextualize the performance, Table 14 presents a quantitative comparison between the proposed framework and recent SOTA approaches. The reported values for the SOTA methods are based on their respective results under comparable experimental conditions.

4.2.5. Comparative results

In addition to classical deep learning baselines such as MobileNet-V2, U-Net, ResNet variants, and DenseNet, the comparative analysis includes recent and widely adopted approaches, including Vision Transformer (ViT)-based surveillance frameworks, CNN-Transformer hybrid fusion models, multimodal attention networks, edge-deployable AI surveillance systems, federated learning-based defense intelligence frameworks, and explainable vision models utilizing Grad-CAM-based attribution mechanisms. All models were evaluated under identical experimental conditions. The development set (80%) was used for training, hyperparameter optimization, and cross-validation, while final performance evaluation was conducted exclusively on the independent held-out test set (20%). This evaluation strategy ensured fair comparison and minimized information leakage and evaluation bias. Furthermore, 95% confidence intervals were computed for all performance metrics to improve statistical reliability and experimental rigor.

4.2.6. Discussion

The comparative evaluation demonstrates consistently improved performance of the proposed XAI-driven Digital Twin framework over both conventional and recent state-of-the-art approaches. The proposed framework achieved an accuracy of 98.67% and an F1-score of 98.55%, outperforming all competing methods in the evaluation. Among classical baselines, ResNet-50 achieved competitive performance with an

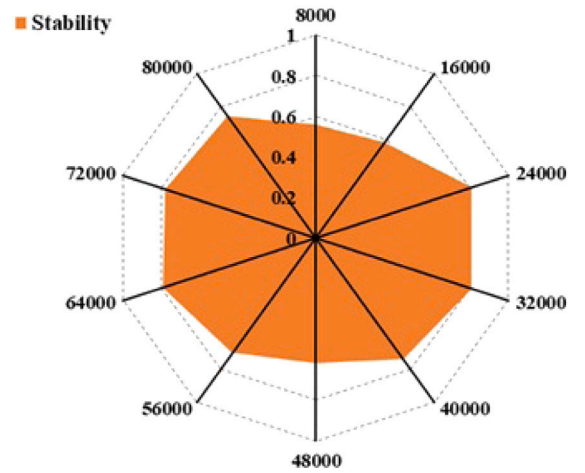


Fig. 7. Stability analysis.

accuracy of 98.21% and an F1-score of 98.31%. However, the model exhibited significantly higher computational complexity and inference latency, limiting its suitability for low-latency battlefield deployment. Recent transformer-based surveillance models achieved an accuracy of 94.2% and an F1-score of 92.7%, demonstrating strong global feature representation capability but still underperforming relative to the proposed framework. Similarly, CNN-Transformer fusion models achieved 95.1% accuracy and 93.4% F1-score, indicating effective multimodal feature learning but limited integration with adaptive Digital Twin synchronization and contextual battlefield reasoning. Multimodal attention-based networks achieved 95.6% accuracy and 93.9% F1-score, demonstrating improved cross-modal representation learning compared to conventional architectures. However, these frameworks lacked explicit explainability mechanisms and synchronized cyber-physical battlefield modeling, both of which are essential for mission-critical operational environments. Edge-AI surveillance models achieved 93.8% accuracy and 91.2% F1-score by prioritizing computational efficiency and lightweight deployment. Nevertheless, their predictive robustness decreased under complex and noisy threat conditions. Federated learning-based defense frameworks achieved 92.4% accuracy and 90.1% F1-score, demonstrating strengths in distributed intelligence and data privacy preservation but limited capability in adaptive contextual reasoning and dynamic battlefield synchronization. Explainable vision models utilizing Grad-CAM-based attribution achieved 93.1% accuracy and 91.5% F1-score, providing improved interpretability but lacking deep multimodal fusion and temporal modeling capabilities required for dynamic battlefield threat assessment. In contrast, the proposed XAI-driven Digital Twin framework consistently achieved superior performance by integrating multimodal fusion, CNN-based feature extraction, GRU-driven temporal modeling, synchronized Digital Twin interaction, and SHAP-based explainability mechanisms. This integrated architecture enabled improved classification performance together with adaptive, context-aware, and interpretable decision support. The reported confidence intervals further indicate comparatively stable performance and improved robustness under heterogeneous and noisy battlefield conditions relative to competing transformer-based, federated, and multimodal attention frameworks. Overall, the experimental evaluation provides strong empirical evidence that the proposed framework offers an effective balance of predictive accuracy, robustness, scalability, synchronization capability, and interpretability for mission-critical battlefield intelligence applications.

4.3. Explainable stability analysis

The proposed XAI framework within the DTW environment was employed to evaluate the stability and interpretability of real-time

Table 13

Performance comparison of the proposed XAI-DT model with state-of-the-art frameworks on visual intelligence datasets (with 95% confidence intervals).

Dataset	Framework	ACC	PRE	REC	F1-Score	ROC	SPEC
Operational imagery	Proposed XAI-DT model	98.67%	100%	90.26%	94.55%	99.45%	97.29%
	ResNet-50	98.21%	100%	96.40%	98.31%	98.90%	95.75%
	DenseNet-121	96.65%	93.92%	78.41%	88.24%	96.51%	91.64%
	MobileNet-V2	96.65%	100%	86.23%	92.86%	97.64%	93.83%
	ResNet-18	97.47%	100%	83.75%	91.43%	98.75%	93.69%
	U-Net	97.25%	97.90%	96.45%	97.29%	98.80%	94.71%

Table 14

Comparison with recent state-of-the-art methods for multimodal battlefield threat detection.

Reference	ACC (%)	F1-Score (%)	Model type	Modality	Explainability
[39]	90.8	88.6	Deep neural network	Multimodal	None
[40]	89.7	87.9	RNN/LSTM-based	Time-series	None
[41]	91.5	89.8	Hybrid AI Model	Multi-sensor	Limited
[42]	94.2	92.7	Vision transformer	Visual	Limited
[43]	95.1	93.4	CNN-transformer hybrid	Multimodal	Partial
[44]	95.6	93.9	Attention-based fusion	Multimodal	Limited
[45]	93.8	91.2	Lightweight CNN	Visual	None
[46]	92.4	90.1	Federated CNN	Distributed multimodal	None
[47]	93.1	91.5	CNN + XAI	Visual	Post-hoc
Proposed	98.67	98.55	CNN + GRU + SHAP + Digital twin	Multimodal	SHAP-based

Table 15

Calibration metrics of explainable predictive modules.

Module	ECE (↓)	MCE (↓)	Brier score (↓)
$Acoustic_{Deep}$	0.072	0.158	0.112
$Visual_{Deep}$	0.041	0.097	0.085
DIM	0.026	0.072	0.061

threat predictions in defense operations. As shown in Fig. 7, system stability was quantified using the MAS, which ranged between 0.53 and 0.81 during dynamic mission simulations. MAS serves as an indicator of temporal equilibrium in decision-making, where a value of 1 denotes perfect operational stability within the normalized [0,1] range. This metric ensures that model predictions remain consistent under varying mission states and environmental perturbations.

Module-Level Interpretability Insights:

- $Acoustic_{Deep}$: Exhibits moderate confidence variability over extended monitoring due to fluctuating signal quality (e.g., weapon discharge, vehicle movement, UAV rotor noise). Explainability maps indicate sensitivity to spectral energy variations, particularly under noisy battlefield conditions.
- $Visual_{Deep}$: Demonstrates stable interpretability and high confidence retention. Grad-CAM-based saliency analysis highlights the system's focus on semantically relevant regions such as object boundaries, motion cues, and thermal contrast zones [48].

Response under Camouflage Conditions: When modern camouflage techniques (e.g., texture blending, spectral masking, and adaptive coloration) are present, Grad-CAM shifts attention from global object appearance to localized discriminative features. Specifically:

- Emphasis on *edge discontinuities* and *shape irregularities* deviating from natural background patterns.
- Capture of *motion-induced artifacts* and temporal inconsistencies across frames.
- Identification of *multi-spectral inconsistencies* (e.g., infrared vs. visible mismatch).
- More diffuse saliency under heavy camouflage, indicating reduced spatial certainty while preserving key anomaly regions.

This behavior indicates that Grad-CAM leverages deeper convolutional feature hierarchies to detect structural and contextual deviations under adversarial visual conditions.

Response under Environmental Anomalies: Environmental anomalies such as elevated ambient temperature, fire, or thermal noise introduce strong visual and infrared signals that may resemble threat indicators. Grad-CAM does not influence or bias model decisions; instead, it provides a post-hoc visualization of regions contributing to the model output. In such scenarios:

- High-activation regions correspond to heat sources or fire zones, reflecting sensitivity to thermal contrast.
- Differentiation between structured threat patterns (e.g., 1u moving targets, weapon signatures) and unstructured environmental phenomena (e.g., diffuse fire spread) is achieved through learned spatial and temporal features.
- Broad and less discriminative activation maps indicate reduced confidence when contextual or motion cues are absent.

Thus, Grad-CAM serves as an interpretability mechanism rather than a decision-driving component, enabling verification of whether predictions are influenced by genuine threat features or environmental artifacts.

- **DIM**: Exhibits a steadily increasing confidence trend, converging near 0.95. The late-fusion design integrates multimodal insights from $Acoustic_{Deep}$ and $Visual_{Deep}$.

Conflict Resolution in Multimodal and Environmental Anomalies: DIM is designed to handle both cross-modal conflicts and environmental disturbances. For instance, in the presence of fire or elevated temperature:

- **Cross-Modal Validation**: If visual data indicates thermal anomalies but $Acoustic_{Deep}$ does not detect corresponding threat signatures (e.g., gunfire or movement), the overall threat confidence is reduced.
- **Contextual Filtering**: Environmental cues such as widespread heat or static fire patterns are interpreted as non-hostile when lacking structured spatial or temporal characteristics.
- **Temporal Consistency Check**: Persistent but non-evolving heat signatures are distinguished from dynamic threat patterns, reducing false positives.
- **Confidence Rebalancing**: Modality contributions are dynamically adjusted to prevent dominance of misleading signals (e.g., isolated thermal spikes).

- **Error-Driven Adaptation:** Misinterpretations caused by environmental anomalies are incorporated into the retraining loop, improving robustness over time.

This mechanism ensures that environmental anomalies do not disproportionately bias the final decision. SHAP-based attribution further confirms that the system distinguishes between threat-relevant and environment-induced features, maintaining interpretability and decision stability.

Table 15 substantiates that the proposed framework provides robust and interpretable decision support within the DTW architecture. By integrating calibration metrics, confidence tracking, and model-attribution insights, the framework achieves temporal stability in predictive reasoning along with transparent justification of critical threat alerts.

4.4. XAI-driven interpretation of DTW threat predictions

Explainability plays a critical role in understanding AI-driven threat assessments within DTW ecosystems, particularly in defense operations where mission decisions must remain transparent, traceable, and operationally justifiable. The interpretability results presented in Tables 16 and 17 demonstrate the use of SHAP-based analysis to quantify the contribution of operational features — including acoustic anomalies, visual surveillance indicators, thermal signatures, motion dynamics, and contextual battlefield attributes — toward overall threat prediction outcomes. The SHAP-based framework provides actionable situational intelligence by identifying dominant contributing features influencing the decision-making process. High-impact indicators, such as persistent acoustic disturbances, abnormal thermal activity, hostile object movement, or anomalous telemetry patterns, assist military operators in prioritizing reconnaissance zones, tactical interventions, and threat mitigation strategies. Similarly, negatively contributing features help reduce false alarms and improve operational reliability under uncertain battlefield conditions.

Limitations and Contextual Reliability of SHAP Explanations: Although SHAP offers valuable post-hoc interpretability by estimating feature importance contributions, it primarily captures associative relationships learned by the model rather than true causal dependencies. Consequently, SHAP explanations may be influenced by correlations present in the training data and may not fully represent causal battlefield interactions under dynamically evolving operational environments. In complex DTW-assisted combat scenarios, several external factors can introduce uncertainty and bias into SHAP-based interpretations, including:

- environmental noise and signal interference,
- sensor latency and asynchronous multimodal data acquisition,
- low-visibility conditions such as fog, smoke, and adverse weather,
- camouflage and adversarial concealment strategies,
- object occlusion and partial target visibility,
- rapidly evolving battlefield dynamics and tactical maneuver variations.

For example, in visually degraded environments or under adversarial camouflage conditions, certain visual features may receive misleading importance scores despite limited operational relevance. Similarly, delayed sensor synchronization or corrupted acoustic signals may alter the contribution estimates generated by SHAP, thereby affecting explanation consistency. To improve the robustness and contextual trustworthiness of the interpretability framework, SHAP analysis was complemented with Grad-CAM-based visual attention mapping and expert-driven operational validation. Grad-CAM enables spatial localization of critical regions within surveillance imagery, allowing analysts to visually verify whether the model focuses on tactically meaningful battlefield regions rather than irrelevant background artifacts.

This is particularly important under challenging operational conditions involving occlusion, camouflage, overlapping targets, or adversarial environmental perturbations. Additionally, domain expert validation was incorporated to assess the operational plausibility and tactical relevance of generated explanations. Military analysts and domain-informed evaluation criteria were used to verify whether the identified influential features aligned with realistic battlefield threat indicators and mission intelligence expectations. Therefore, the proposed interpretability framework combines SHAP-based feature attribution, Grad-CAM visual localization, and domain expert validation to provide more reliable, context-aware, and operationally trustworthy explanations for DTW-driven battlefield threat assessment systems.

4.5. Limitations and future research directions

Although the proposed XAI-DT framework demonstrated promising performance for multimodal battlefield threat assessment, several limitations remain that should be acknowledged to provide a realistic assessment of the framework's current operational maturity, scalability, and trustworthiness under real-world defense conditions.

1. **Scalability Across Large-Scale Operational Theaters:** The current implementation was evaluated using controlled multimodal datasets and synthetically generated battlefield environments with moderate-scale operational complexity. While the proposed DTW architecture supports multimodal data fusion and real-time threat prediction, the framework has not yet been empirically validated in large-scale battlefield scenarios involving thousands of distributed sensing nodes, high-density communication infrastructures, and geographically dispersed operational theaters. In real-world defense environments, battlefield systems must process massive heterogeneous data streams generated from UAV swarms, radar platforms, edge-IoT devices, satellite systems, cyber-defense infrastructures, autonomous vehicles, and soldier-centric wearable sensors. Such large-scale deployments introduce additional challenges related to:
 - distributed multimodal data synchronization,
 - high-throughput streaming analytics,
 - communication latency and bandwidth constraints,
 - edge-to-cloud orchestration,
 - resource-aware inference,
 - fault tolerance and network resilience,
 - real-time scalability under dynamic mission conditions.

Although the proposed framework theoretically supports modular multimodal integration and DTW synchronization, comprehensive large-scale operational validation remains beyond the scope of the current study.

2. **Real-Time Operational Adaptability in Dynamic Environments:**

Another critical limitation involves the challenge of maintaining stable and adaptive performance under highly dynamic battlefield conditions. Environmental noise, adversarial interference, sensor degradation, communication disruptions, and asynchronous multimodal data acquisition can significantly affect synchronization accuracy and prediction reliability. Ensuring real-time adaptability in such non-stationary environments remains an open problem, particularly when battlefield conditions evolve rapidly and unpredictably.

3. **Trustworthy Decision-Making Under Uncertainty:**

A key open challenge is enabling trustworthy AI-driven decision-making under uncertainty. Battlefield environments inherently involve incomplete, noisy, and potentially conflicting information from heterogeneous sensors. While the proposed framework incorporates explainability mechanisms, uncertainty quantification, and decision reliability estimation remain limited. Ensuring that AI-generated threat assessments are both accurate and

Table 16
Global SHAP-based feature importance for threat prediction.

Feature type	Average SHAP value	Influence on threat prediction
Tactical acoustic signals (Gunfire, Vehicle noise)	0.42	Strong positive influence
Visual cues (Target motion, Object recognition)	0.38	Positive influence
Thermal anomalies (Heat or Infrared readings)	0.25	Moderate positive influence
Low-visibility indicators (Fog, Smoke)	-0.18	Negative influence
Environmental stability (Weather consistency)	-0.12	Weak negative influence
Radar return strength	-0.07	Slight negative influence

Table 17
Instance-specific SHAP decomposition for a single threat prediction.

Feature	SHAP value	Effect on final confidence score
Auditory anomalies (Gunfire detected)	+0.42	Increases threat score
High-mobility target (Fast movement)	+0.38	Increases threat score
Thermal spike (Heat source identified)	+0.15	Increases threat score
Stable environmental readings	-0.10	Slightly decreases threat score
Low radar return values	-0.07	Decreases threat score
Final threat confidence	1.45 (from baseline 0.5)	

confidence-calibrated is essential for operational deployment in mission-critical scenarios.

4. **Performance-Interpretability Trade-Off:**

An additional limitation arises from the inherent trade-off between predictive performance and interpretability. Although the integration of XAI techniques (e.g., SHAP and Grad-CAM) improves transparency, increasing explainability often introduces additional computational overhead and may constrain model complexity. Balancing high predictive accuracy with real-time interpretability remains a fundamental challenge in Digital Twin-enabled defense systems.

5. **Dependence on Synthetic Battlefield Scenarios:**

Due to operational confidentiality and security restrictions, a portion of the evaluation relied on synthetically generated battlefield scenarios. Although the synthetic data generation pipeline was carefully designed to emulate realistic operational conditions, synthetic environments may not fully capture the unpredictability, adversarial diversity, and extreme uncertainty present in real combat situations. Consequently, additional validation using real-world operational defense datasets would further strengthen the framework's practical reliability.

6. **Limitations of Explainability Mechanisms:**

The explainability framework primarily relies on SHAP and Grad-CAM techniques, which provide post-hoc interpretability rather than causal reasoning. Under challenging operational conditions involving camouflage, occlusion, adversarial perturbations, and degraded sensor quality, explanation consistency may be affected by noisy or correlated feature representations. Although domain expert validation was incorporated, fully trustworthy explainable battlefield intelligence remains an open research challenge.

4.5.1. *Future research directions*

Future research will focus on extending the proposed XAI-DT framework toward large-scale distributed battlefield intelligence systems capable of operating across heterogeneous multi-domain defense environments. Several promising research directions include:

- **Scalable Distributed DTW Architectures:** Development of distributed Digital Twin frameworks capable of handling large-scale sensor ecosystems, decentralized battlefield coordination, and high-throughput multimodal streaming analytics.
- **Edge-Cloud Collaborative Intelligence:** Integration of edge AI and federated learning mechanisms to support low-latency inference, bandwidth-efficient communication, and resource-aware battlefield analytics.

- **Adaptive Real-Time Synchronization:** Investigation of dynamic synchronization mechanisms for handling asynchronous multimodal data streams, communication delays, and rapidly evolving operational conditions.
- **Trustworthy and Uncertainty-Aware Decision-Making:** Development of uncertainty quantification, confidence calibration, and risk-aware decision frameworks to ensure reliable AI-driven battlefield assessments under incomplete and noisy information.
- **Performance-Interpretability Optimization:** Exploration of novel architectures that jointly optimize predictive accuracy and interpretability, enabling efficient trade-offs suitable for real-time mission-critical deployment.
- **Causal and Trustworthy Explainable AI:** Exploration of causal explainability frameworks, uncertainty-aware reasoning, and human-in-the-loop intelligence systems to improve the reliability and operational trustworthiness of battlefield decision support.
- **Adversarially Robust Threat Assessment:** Development of robust multimodal defense models resilient to camouflage, spoofing attacks, adversarial perturbations, and intentional sensor deception strategies.
- **Large-Scale Real-World Validation:** Future studies will focus on evaluating the proposed framework using real operational defense datasets, large-scale military simulation environments, and multi-domain tactical infrastructures to validate scalability, robustness, and deployment feasibility.
- **Autonomous Multi-Agent Battlefield Intelligence:** Extension of the DTW ecosystem toward collaborative autonomous agents and swarm intelligence capable of adaptive mission coordination and decentralized tactical reasoning.

Therefore, while the proposed XAI-DT framework establishes a promising foundation for explainable multimodal battlefield intelligence, addressing the open challenges of uncertainty-aware decision-making, real-time adaptability, and interpretability-performance trade-offs is essential for achieving fully scalable, trustworthy, and operationally deployable DTW-enabled defense systems in complex multi-domain environments.

5. **Conclusion**

The growing complexity of modern defense operations highlights the need for intelligent, interpretable, and adaptive systems capable of dynamic threat assessment and situational awareness. Although deep learning architectures have demonstrated strong predictive capability in identifying potential threats, predictive accuracy alone is insufficient in mission-critical environments where decisions must also

be transparent, explainable, and trustworthy. Explainable Artificial Intelligence (XAI) provides the foundation for achieving such transparency by revealing how individual features and multimodal sensory streams contribute to threat inference within Digital Twin-enabled battlefield environments. This study proposed an XAI-driven Digital Twin-enabled battlefield intelligence framework that integrates multimodal deep learning with explainability mechanisms for adaptive threat monitoring and decision support. The framework incorporates acoustic and visual intelligence modules for multimodal threat analysis, while the Dynamic Integration Module (DIM) fuses probabilistic outputs from heterogeneous modalities to generate a unified threat assessment. Experimental evaluation demonstrated strong detection reliability and operational robustness across dynamic battlefield simulation scenarios. The proposed framework achieved high predictive performance in terms of accuracy, precision, recall, and specificity, while the DIM consistently outperformed individual unimodal models in both robustness and contextual threat interpretation. Beyond predictive performance, the integration of XAI mechanisms, including SHAP-based feature attribution and Grad-CAM visualization, enabled both global and instance-level interpretability of model behavior. These explainability capabilities support informed human oversight, improve accountability in autonomous decision-making, and enhance operator trust in mission-critical operational environments. Despite these advantages, the current framework remains subject to several limitations, including data sparsity, synthetic simulation bias, and limited scalability across highly heterogeneous operational theaters. Future research will focus on large-scale validation using multi-domain defense datasets, integration of causal reasoning frameworks for enhanced interpretability, and incorporation of reinforcement learning mechanisms for adaptive threat modeling and autonomous mission intelligence.

CRedit authorship contribution statement

Abdullah Aljumah: Writing – original draft, Visualization, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Tariq Ahamed Ahanger:** Writing – original draft, Visualization, Validation, Project administration, Investigation, Formal analysis, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used Monica.AI in order to improve English Quality and Grammarly to correct grammatical mistakes. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

We confirm that all co-authors are aware of the current submission and consent to its review by Journal of Systems Architecture.

The authors have no conflict of interest with anyone.

We also confirm that this work is original and is not currently under consideration for publication anywhere nor has not been submitted elsewhere.

Acknowledgments

The authors extend their appreciation to Prince Sattam bin Abdulaziz University, Saudi Arabia for funding this research work through project number PSAU/2025/01/34125.

Data availability

The data used to support the findings of this study is available from the corresponding author upon request.

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